Marine Microbial Biodiversity

Lecture 07

Bacterial and Archaeal Diversity in Marine Environments

Diversity 2

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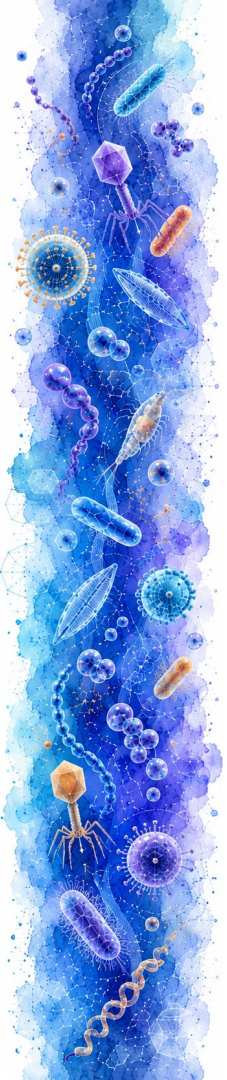
@matteo.selci

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Self Evaluation Test

When?

. November 18, 9:30am → First 10 lectures

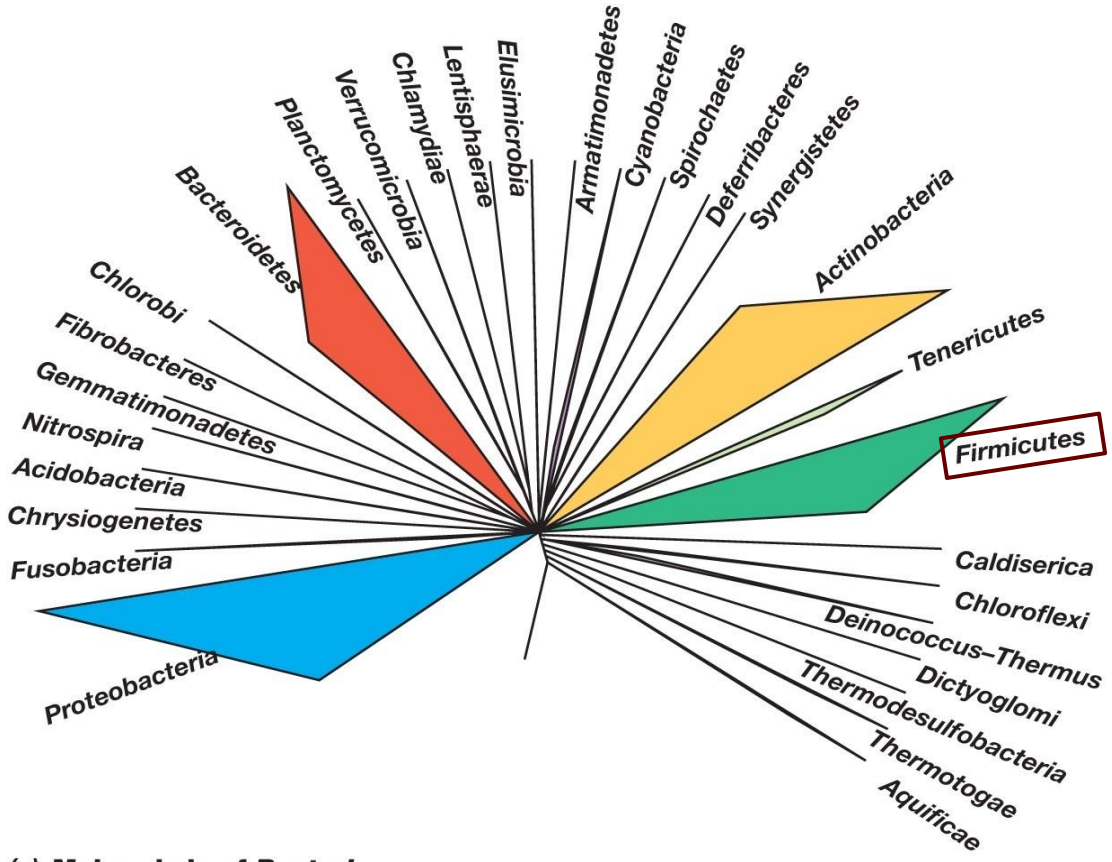
. December 16, 9:30am → Remaining lectures

How?

. 30 multiple choice questions in 30 minutes

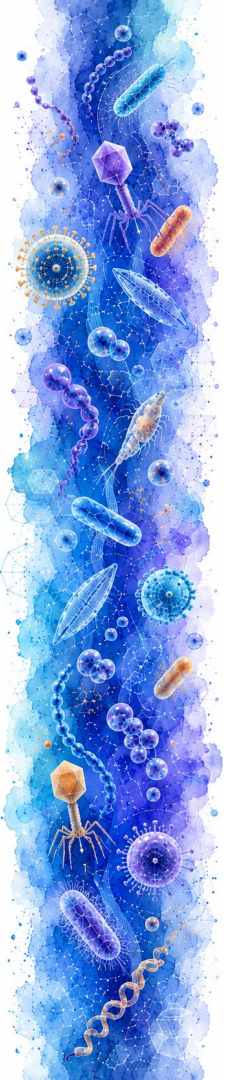
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The Tree Of Life: Bacteria



(a) **Major phyla of *Bacteria***

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Bacterial Phyla: Firmicutes

Firmicutes (now Bacillota) represent one of the two major subdivisions of Gram-positive bacteria. They are also known as the low-G+C gram-positives, typically having a cell wall consisting of a thick layer of peptidoglycan, while a few rather unusual genera lack a cell wall entirely.

They generally have round cells (**cocci**) or rod-like forms (**bacilli**). They are found in various environments, including sediments, hydrothermal vents, and the open ocean. They play essential roles in organic matter degradation and biogeochemical cycling, especially under anaerobic conditions.

Firmicutes are generally chemoorganotrophic and/or fermenters. Many of them are also known to produce endospores in response to starvation or harsh chemical or physical conditions, and can survive extreme conditions. Whereas Heliobacteriaceae are the only family belonging to this phylum which is able to produce energy through anoxygenic photosynthesis.

The group is typically divided into the Clostridia, which are obligate anaerobes, and the Bacilli, which are obligate or facultative aerobes.

Bacillus species are almost ubiquitous in nature, e.g. in soil, but also occur in extreme environments such as high pH (*B. alcalophilus*), high temperature (*B. thermophilus*), or high salt (*B. halodurans*)

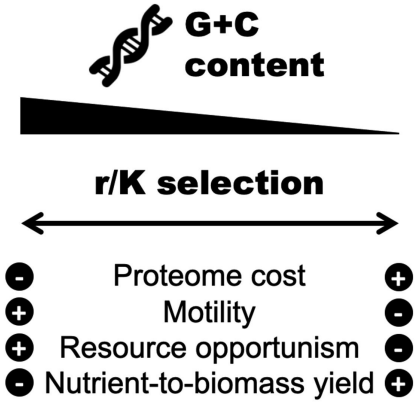
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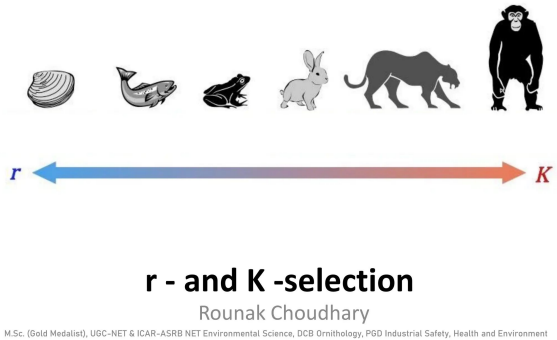
Bacterial Phyla: Firmicutes

Low G+C content aligns with a specialized or stress-tolerant lifestyle, often linked to distinct metabolic trade-offs, endospore formation, and survival in fluctuating or nutrient-poor niches.

This is representative of K-strategists that are specialized in stable environments in which they maintain resource efficiency, competitive specialism, and high survival rates.

On the opposite side, few rather unusual Firmicutes genera lack a cell wall entirely such as Mycoplasma and Phytoplasma. They have adapted to different lifestyles like obligate commensalism or pathogenicity, living inside or strictly attached to the mucosal surfaces of eukaryotic hosts.





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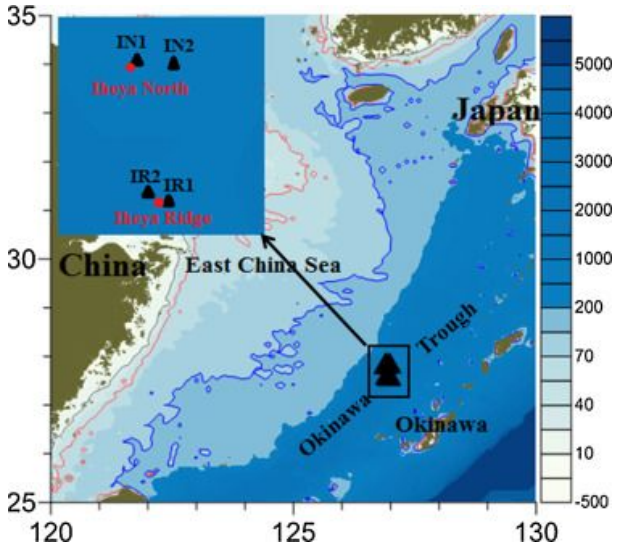
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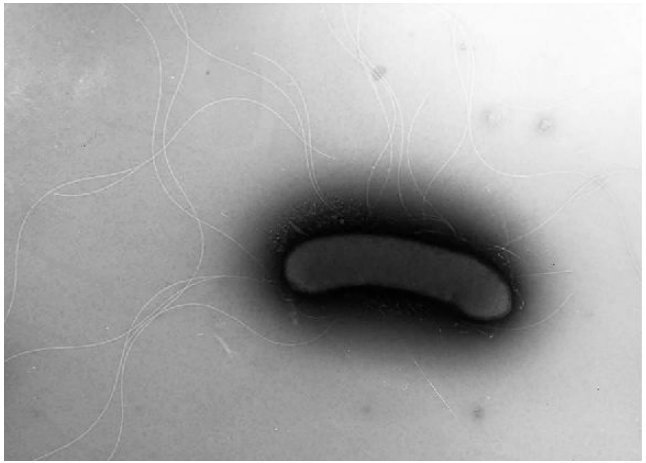
Bacterial Phyla: Firmicutes

Oceanobacillus iheyensis HTE831 is an extremely **halotolerant** and **alkaliphilic bacterium** that belongs to the phylum Firmicutes, and more specifically to the family Bacillaceae.

It is an **obligate aerobe heterotroph**, participating in the degradation of organic matter in saline and alkaline marine environments, though specific enzymatic activities.
It was isolated from deep sea sediment collected at a depth of 1050 m on the Iheya Ridge, a volcanic zone near Okinawa.

The genome of *O. iheyensis* is a single circular chromosome consisting of 3,630,528 nt
O. iheyensis has at least 29 hydrolytic enzymes, all of huge interest because they can remain stable and active in high-salt and high-pH conditions, which could be important in industrial applications such as production of detergents, leather processing, or bioremediation of saline environments.

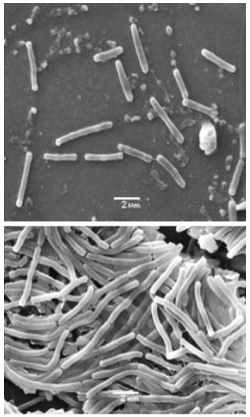


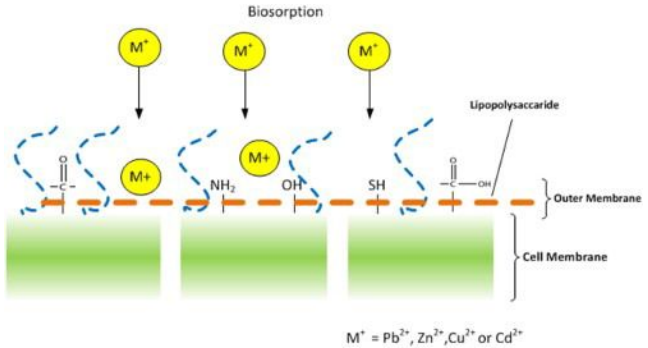


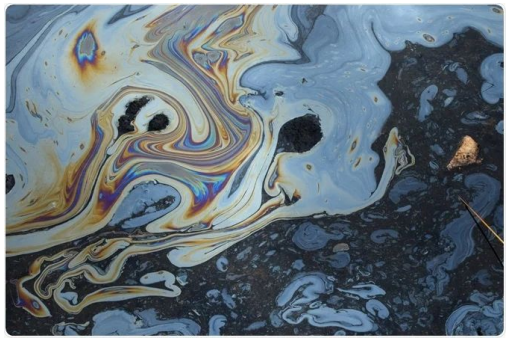
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Bacterial Phyla: Firmicutes

Many have **biotechnological applications**. For example, some Bacillus species bind and oxidize metals such as manganese and copper through a process called **biosorption**, and the spores may have applications in metal **bioremediation** of polluted marine sediments.
Bacillus species are frequently used in **hydrocarbon bioremediation strategies** for oil spill cleanups due to their ability to degrade hydrocarbons and produce biosurfactants.
Some members of the related genus Paenibacillus produce powerful chitinolytic activity, which may be useful in **waste management** for the breakdown of shrimp and crab shell material.



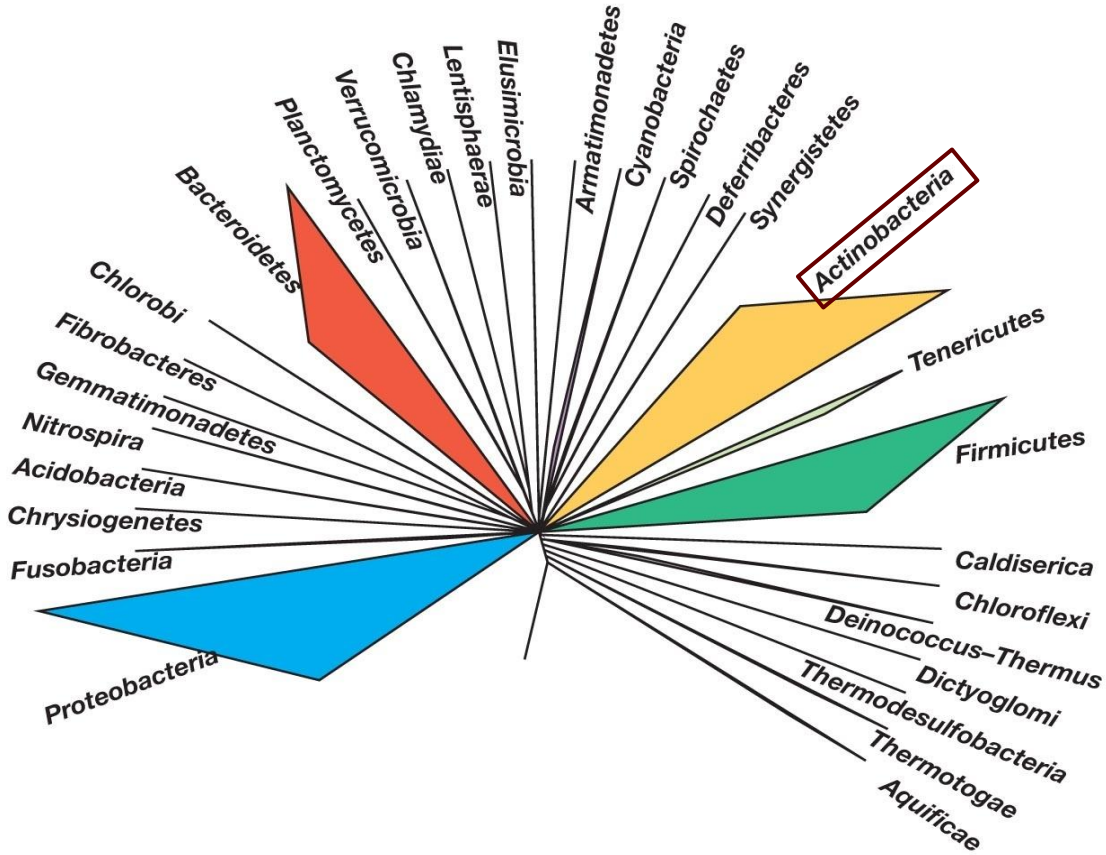






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The Tree Of Life: Bacteria



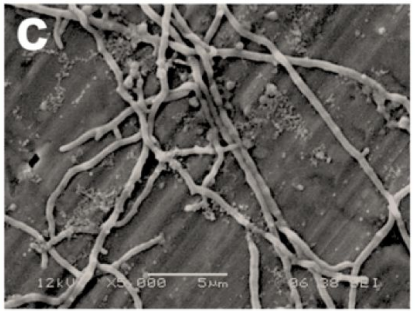
(a) **Major phyla of *Bacteria***

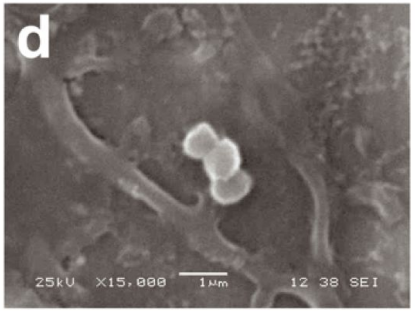
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Bacterial phyla: Actinobacteria

Actinobacteria (now Actinomycetota), in contrast to the Firmicutes, are a phylum of bacteria whose genomic DNA generally have a high GC content, and as a result, they are also called the the high-G+C gram-positives.
Most Actinobacteria are aerobic. Further, they can be chemoautotrophic, but most are chemoheterotrophs able to use a wide variety of nutritional sources, including various complex polysaccharides
Actinomycetes are filamentous bacteria that share morphological similarities with fungi. Many are symbiotic, living in association with marine animals like sponges, corals, and even fish. In these symbiotic relationships, Actinobacteria can produce bioactive compounds that protect their hosts from pathogens or contribute to the host's nutrient acquisition.







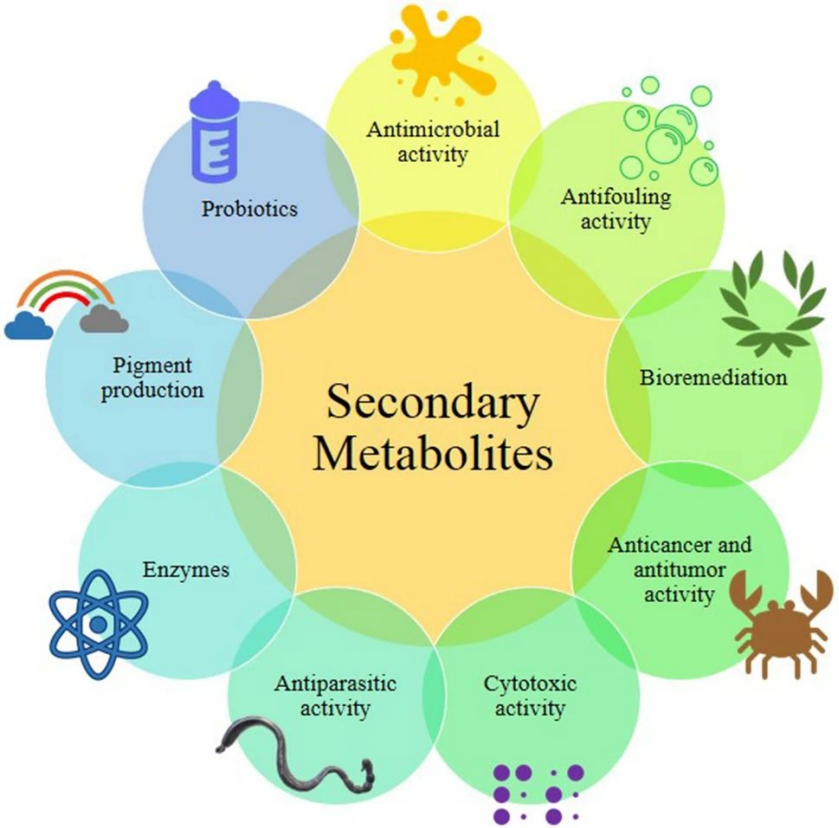
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Bacterial phyla: Actinobacteria

Actinomycetales order is widely distributed in marine sediments and particulate matter in the water column. Because of their high abundance in soil, it is possible that many actinomycetes in coastal sediments are derived from terrestrial runoff, but there are also halophilic species found in deep-sea samples.

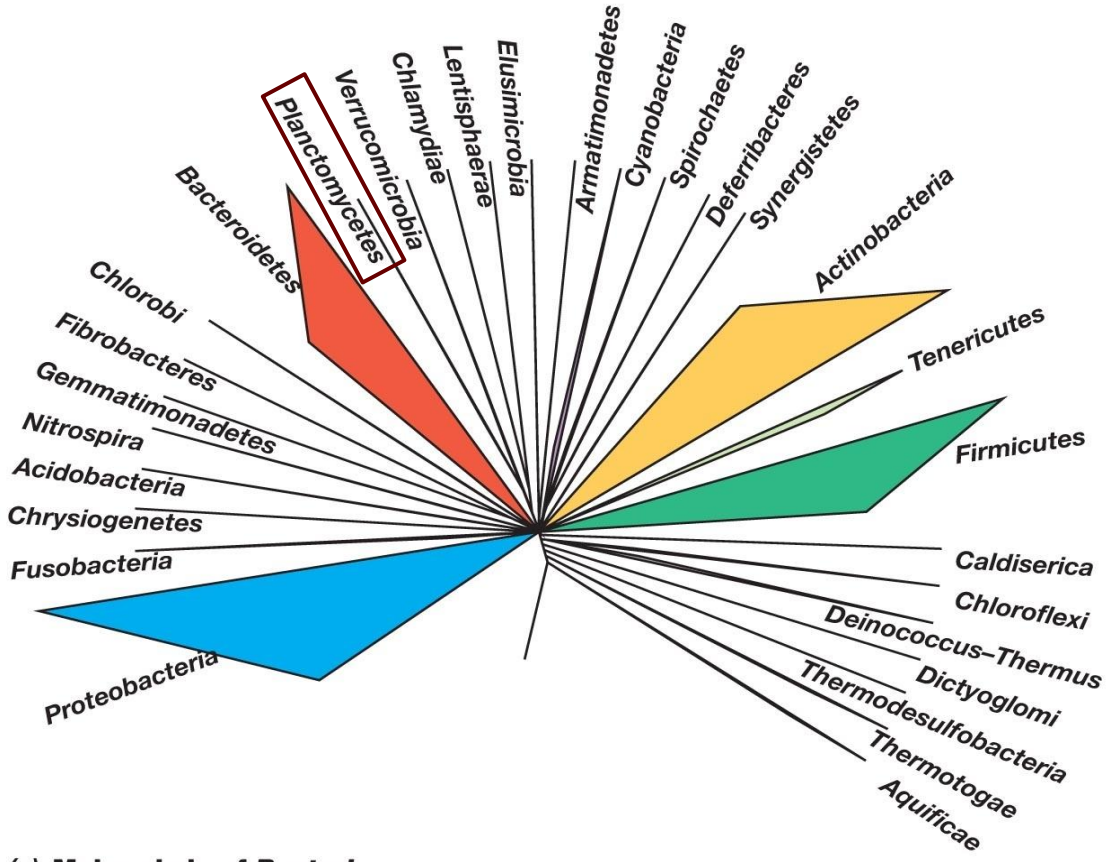
Their main ecological importance is in decomposition and heterotrophic nutrient cycling, owing to their production of diverse extracellular enzymes, which break down polysaccharides, proteins, and fats.

They represent an exceptionally rich source of secondary metabolites. In fact, the most widely used antibiotics (e.g. aminoglycosides and tetracyclines) come from Streptomyces, one of the largest bacterial genera belonging to Actinomycetes.



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The Tree Of Life: Bacteria



(a) **Major phyla of *Bacteria***

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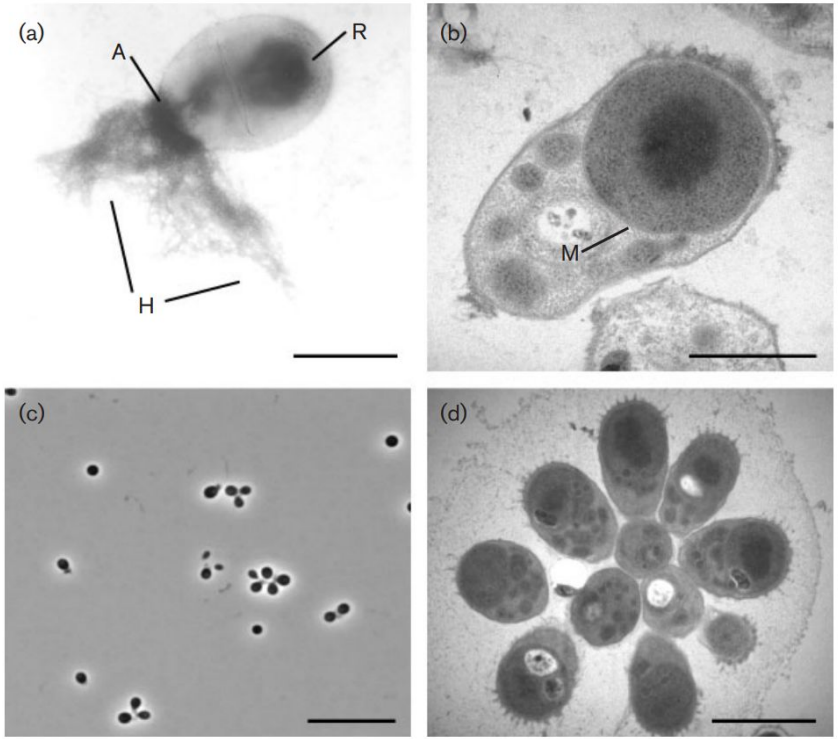
Bacterial phyla: Planctomycetes

Planctomycetes are a phylum of widely distributed bacteria, occurring in both aquatic and terrestrial habitats.

They aerobic chemoheterotrophs and can be grown on common organic media, although many grow very slowly, often living in marine sediments, surfaces, and seawater.

They are likely to play a major role in global carbon cycle, throughout complex organic matter degradation in sediments, and in nitrogen cycle, carrying out anammox metabolism.

They have unusual features of cell structure and morphology, divide by polar budding and are often attached to surfaces, or each others, in rosette-like structures



Rhodopirellula baltica

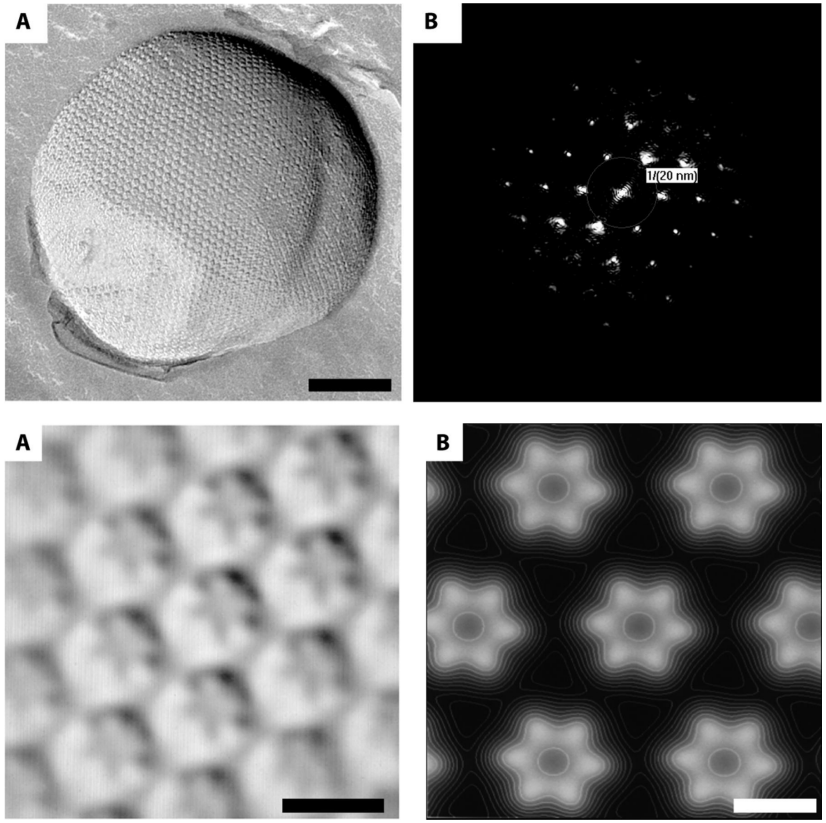
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10.1099/IJS.0.63113-0

Planctomycetes: *Ca. Kuenenia stuttgartiensis*

Candidatus Kuenenia stuttgartiensis is one of the most well-studied and significant species of **anammox bacteria**, belonging to the Planctomycetes phylum. In marine environment it plays a crucial role in the nitrogen cycle since throughout anaerobic ammonium oxidation **remove ammonium and nitrite from ecosystems, converting them into nitrogen gas (N_2)**, and for this can be applied in wastewater treatment.

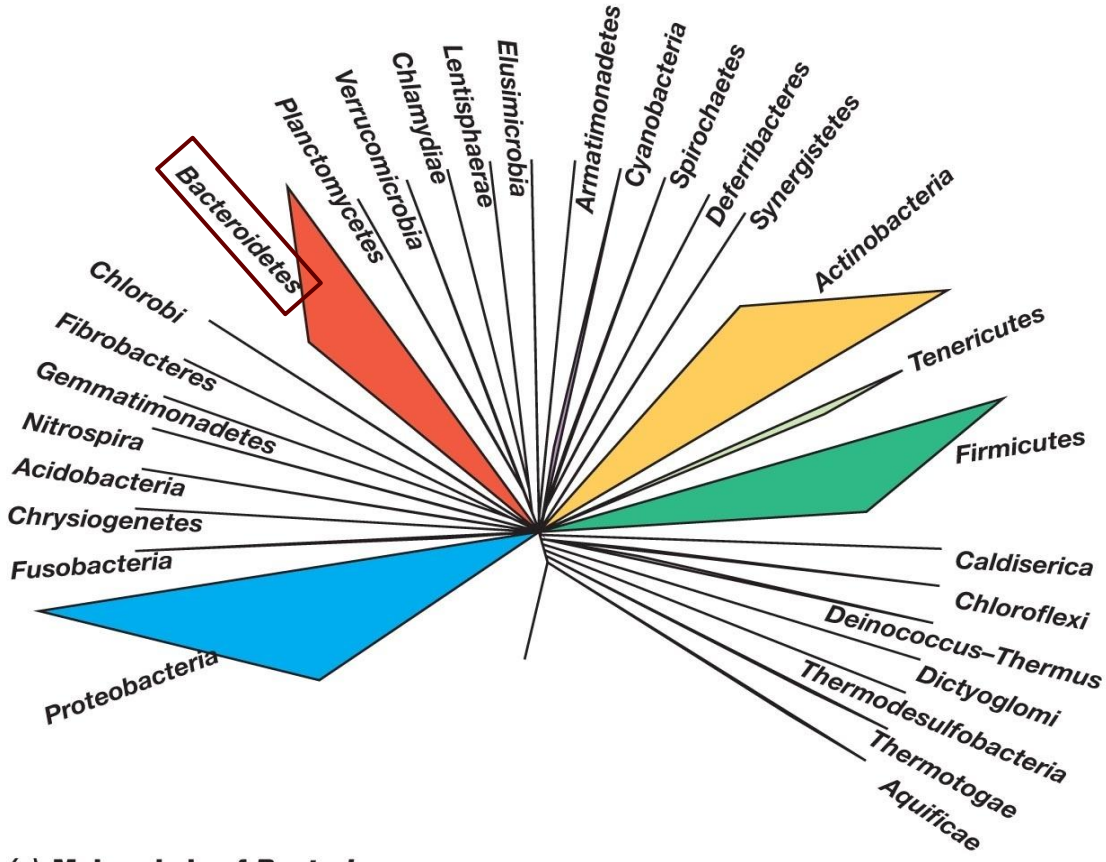
The cell envelope of this bacterium has been proposed to show a proteinaceous surface layer (S-layer) as the outermost component, endowed with a peculiar hexagonal symmetry. Its main component seems to be a glycosylated six subunits protein which has no homology to any known protein. It maintains cell shape, shields against host immune responses or toxins, and mediates surface adhesion.



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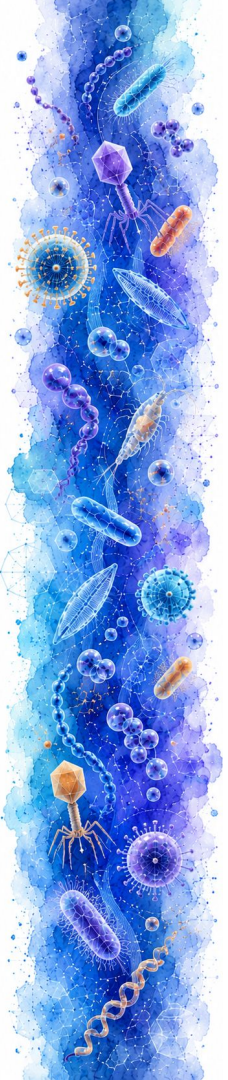
10.1128/jb.00988-13

The Tree Of Life: Bacteria



(a) **Major phyla of *Bacteria***

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Bacterial phyla: Bacteroidetes

The phylum Bacteroidetes (known as the Cytophaga, Flavobacterium and Bacteroides CFB group) is composed of three large classes of non-spore forming, anaerobic or aerobic heterotrophs, and rod-shaped bacteria that are widely distributed in the environment, including in soil, sediments, and sea water, as well as in the guts and on the skin of animals comprising **7000 different species.**

In the marine environment they are ubiquitous. They produce a wide range of hydrolytic extracellular enzymes, which are responsible for degradation of carbohydrate polymers, chitin, and proteins, playing a major role in degrading organic matter, and decomposing complex recalcitrant compounds.

So far, the ones in the Bacteroidia class are the most well-studied, including the genus Bacteroides (an abundant organism in the feces of warm-blooded animals including humans), and Porphyromonas, a group of organisms inhabiting the human oral cavity. Yet, the marine diversity has been poorly investigated.

Many marine isolates of the genera Cytophaga, Cellulophaga, and Flavobacterium have **unusual** flexirubin and carotenoid **pigments** and can be easily isolated as colored colonies on agar media inoculated from sediments, marine snow, and the surfaces of animals and plants and incubated aerobically at ambient temperatures.

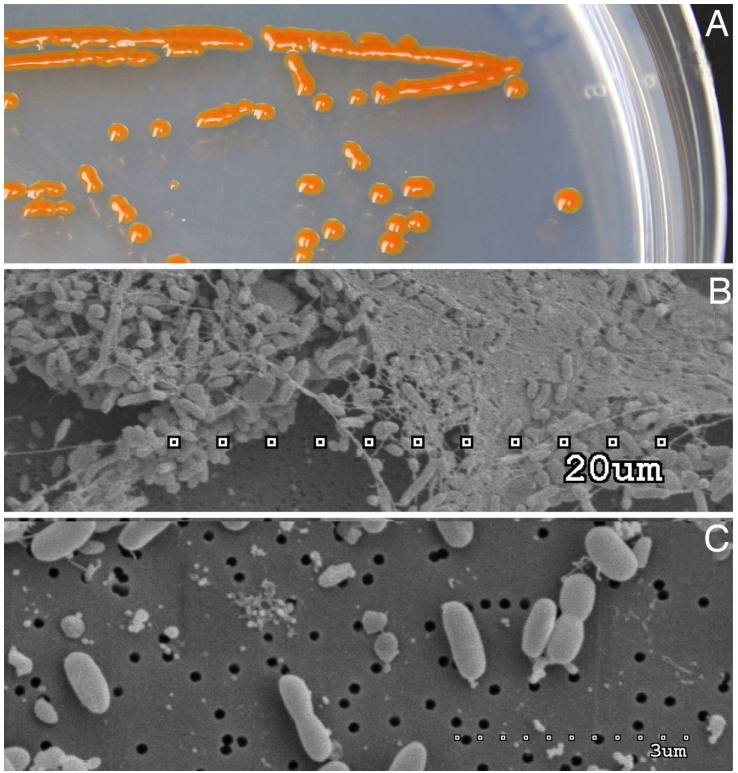
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Bacterial phyla: Bacteroidetes

Polaribacter is one of the major genera of Bacteroidetes found in the marine environment, prominent in **polar habitats or colder temperate regions and deep-sea environments**, where they degrade dissolved organic matter (DOM) and contribute to carbon cycling under extreme conditions.

They contain a proteorhodopsin, acting as a light-driven proton pump; it captures light energy and uses it to move protons across the membrane out of the cell (photoheterotroph). The resulting proton gradient is subsequently used to generate ATP, helping them in oligotrophic environments.

Polaribacter *sp.* MED152 can grow only by aerobic heterotrophy, and it is unable to use fermentation or anaerobic respiration for energy conservation
Its genome contains a number of genes potentially involved in anaplerotic metabolism (fixation of CO_2) such as PEP carboxylase and pyruvate carboxylase



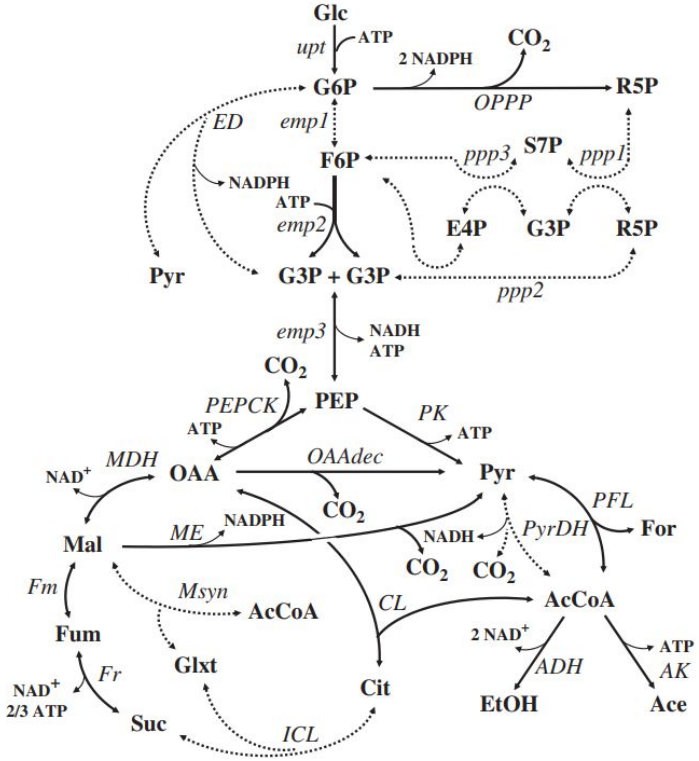
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10.1073/pnas.0712027105

Anaplerosis

Anaplerotic reactions, a term coined by Hans Kornberg and originating from the Greek ἀνά= 'up' and πληρόω= 'to fill', are chemical reactions that form intermediates of a metabolic pathway. Examples of such are found in the citric acid cycle (TCA cycle).

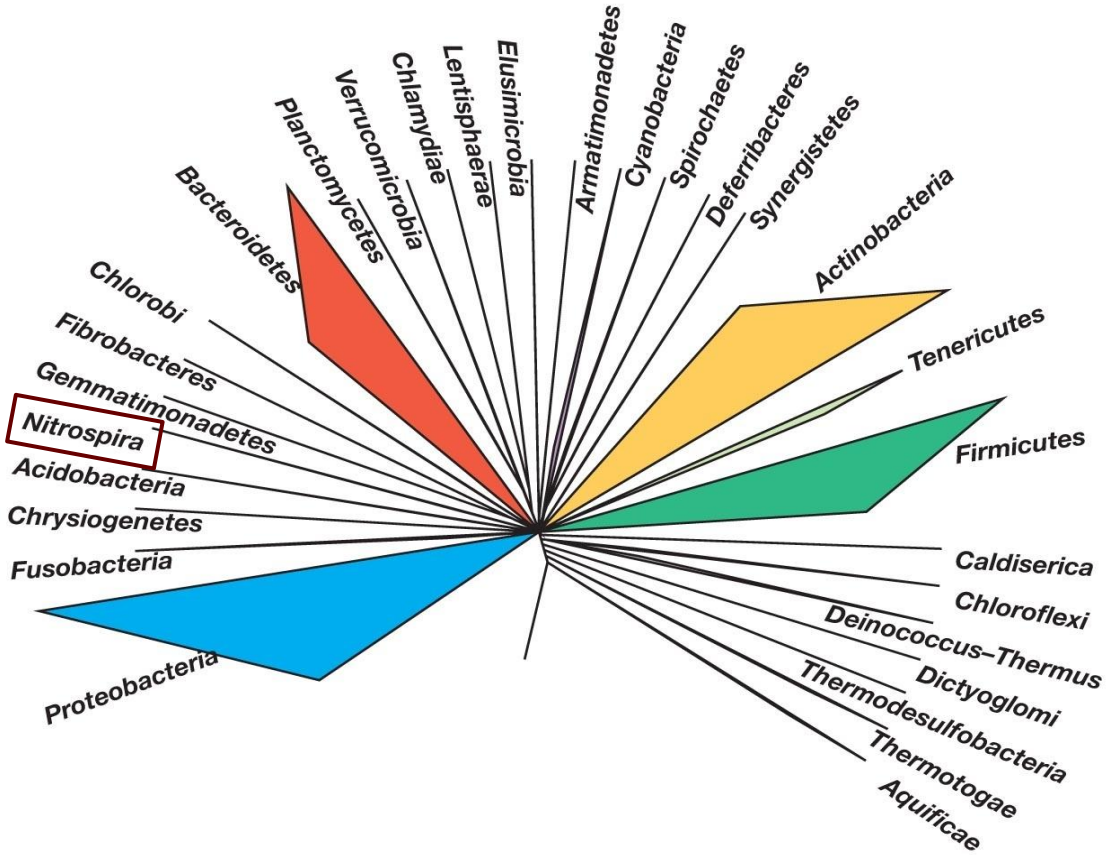
Actinobacillus succinogenes (Gamma proteobacterium) have developed a phosphoenolpyruvate carboxykinase (PEPCK) that converts phosphoenolpyruvate (PEP) directly to oxaloacetate. It acts as anaplerotic enzyme, replenishing a TCA cycle intermediate, but in an energetically favourable way, since the carboxylation of PEP to OAA occurs with the simultaneous generation of ATP.



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10.1073/pnas.0712027105

The Tree Of Life: Bacteria

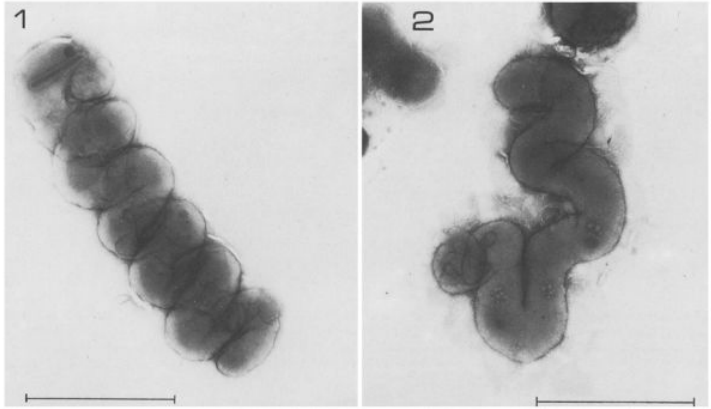


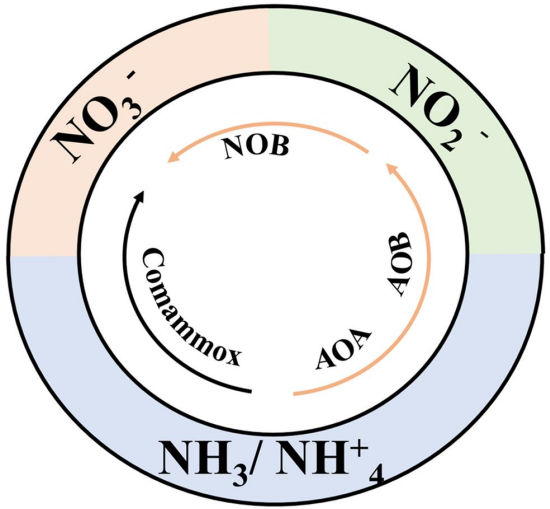
(a) **Major phyla of *Bacteria***

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Bacterial phyla: Nitrospirae

Nitrospirae is a phylum of bacteria containing only one class, Nitrospira, which itself contains one order (Nitrospirales) and one family (Nitrospiraceae).
They are using the **rTCA cycle to fix CO_2 and oxidizing nitrite to nitrate**. They likely represent major nitrifiers in marine ecosystems.
Recently, members of the bacterial genus Nitrospira have been shown to be capable of complete ammonia oxidation ("comammox") to nitrate.
Despite being abundant in many natural and engineered ecosystems Nitrospira are difficult to culture, so most knowledge of them is from molecular and genomic data.
First characterized in 1986, Nitrospira marina is a Gram-negative curved rod which may form spirals with 1 to 12 turns.



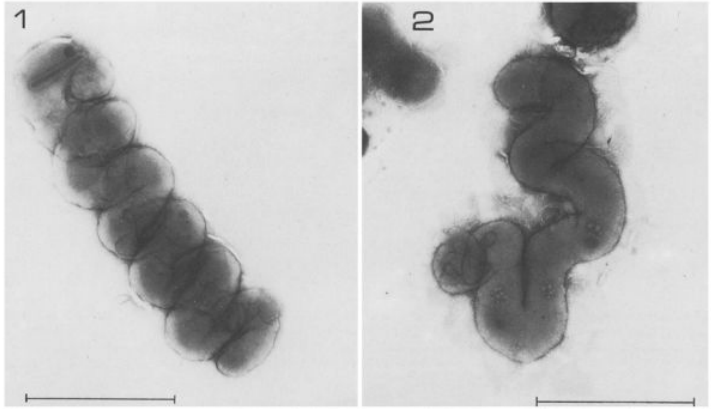


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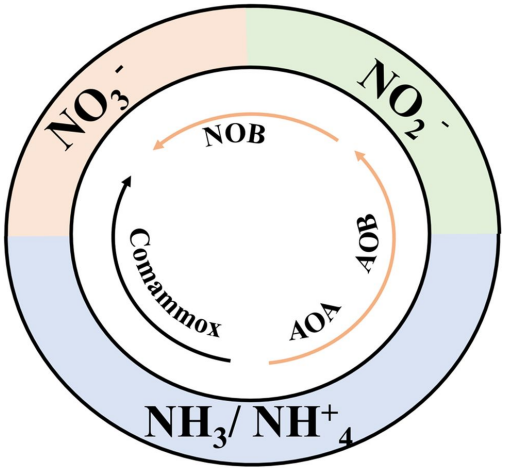
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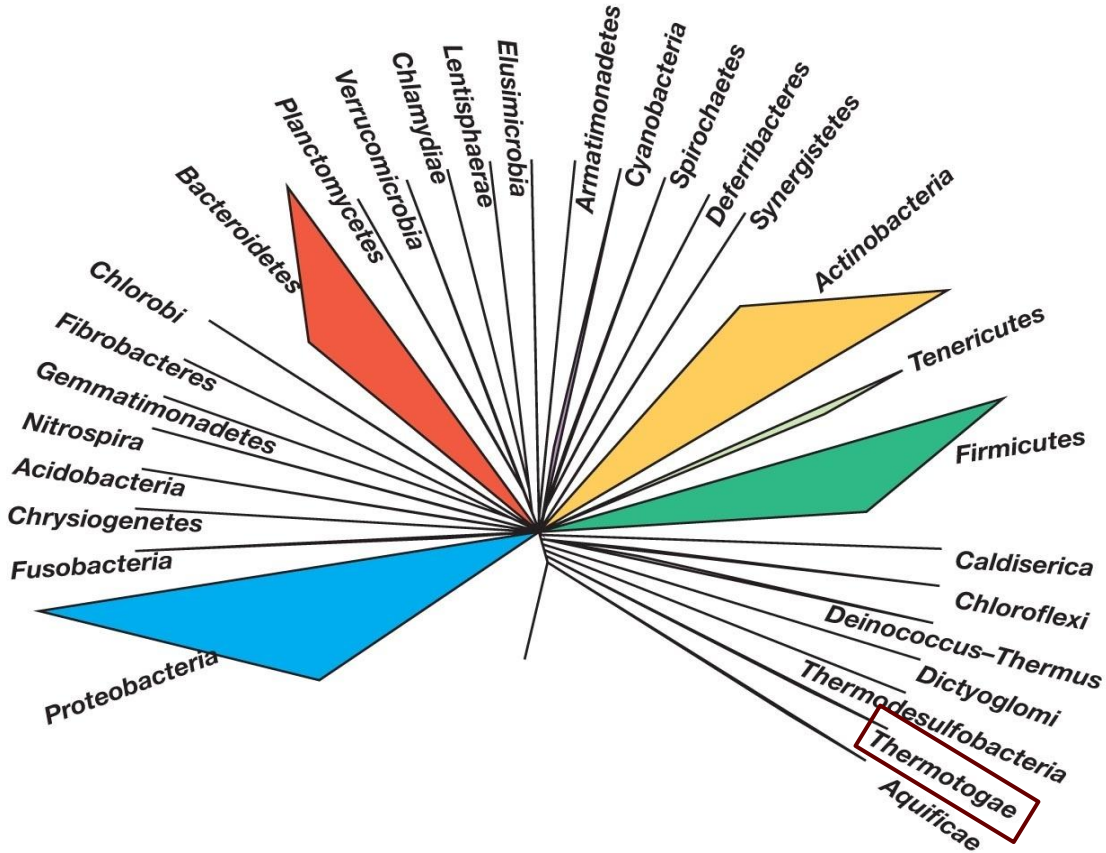
Nitrospira marina



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10.1007/BF00454947

The Tree Of Life: Bacteria



(a) **Major phyla of *Bacteria***

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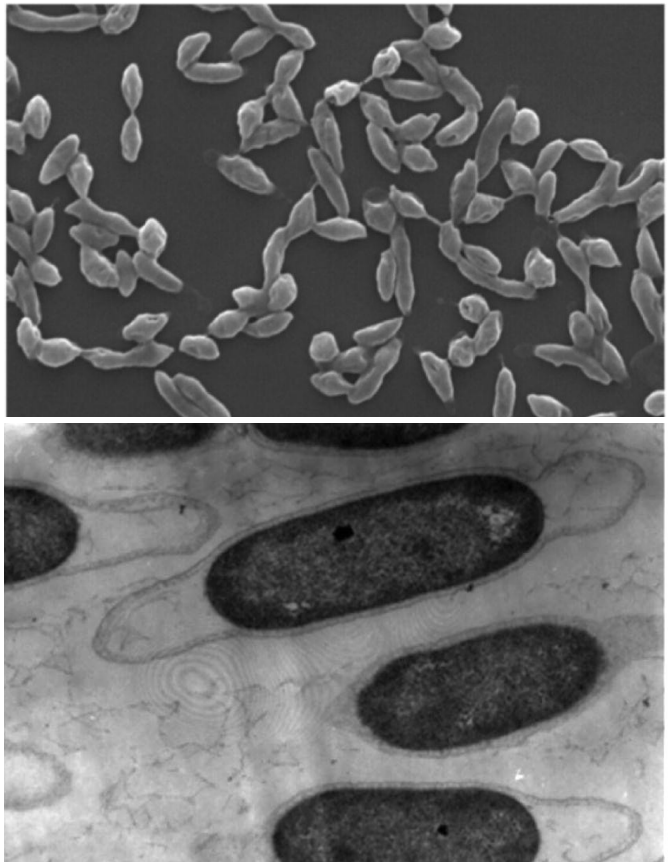
Bacterial phyla: Thermotogae

Thermotogae phylum forms a deeply branching, phylogenetically distant group of gram-negative bacteria. The name of the genus derives from a unique outer membrane ("toga"), which bulges out from the rod-shaped cells.

They are fermentative, anaerobic chemoorganotrophs and use a wide range of carbohydrates. They can also fix N_2 and reduce sulfur to H_2S. They can also grow by anaerobic respiration using H_2 as an electron donor and ferric iron (Fe^{3+}) as acceptor.

Over 20% of the genes of Thermotoga genus probably (?) originated from Archaea by horizontal gene transfers.

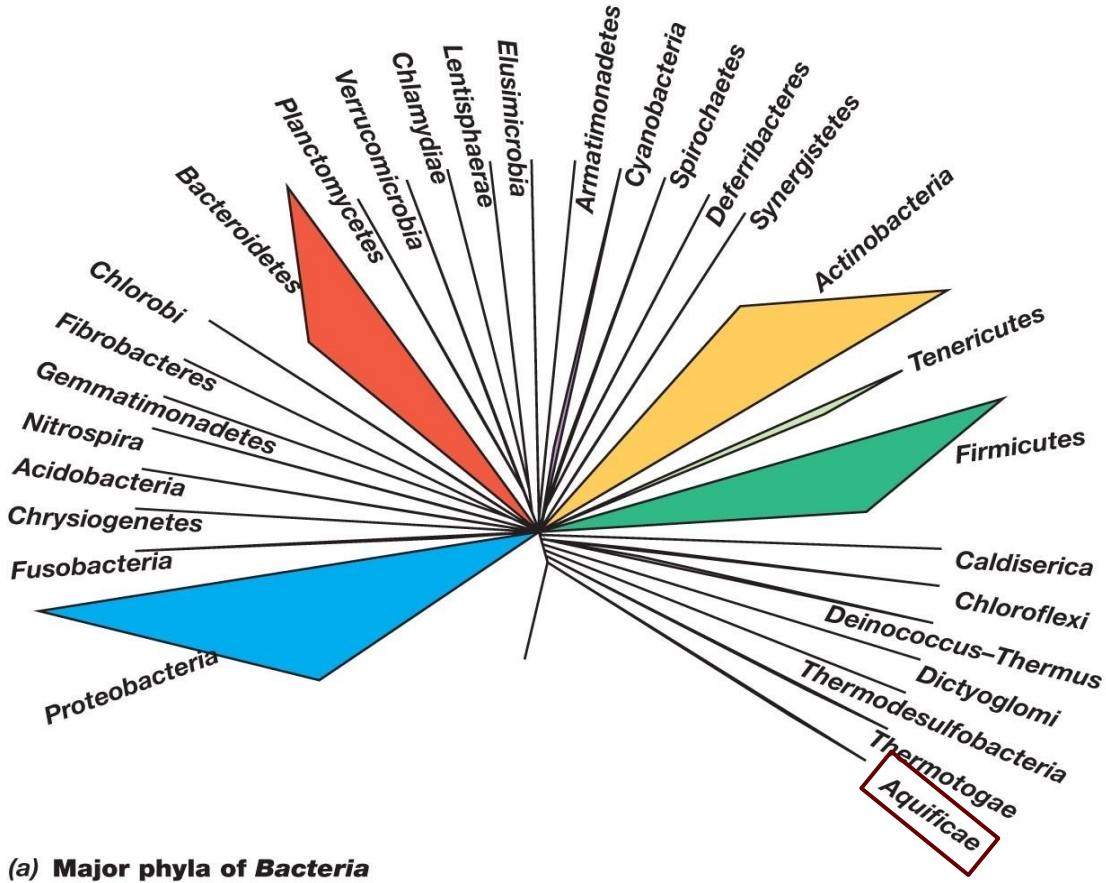
T. maritima and *T. napolitana* are hyperthermophilic species isolated from hydrothermal vents, growing at temperatures ranging from 55°C up to 80 - 95°C



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10.1016/j.tim.2022.09.009

The Tree Of Life: Bacteria



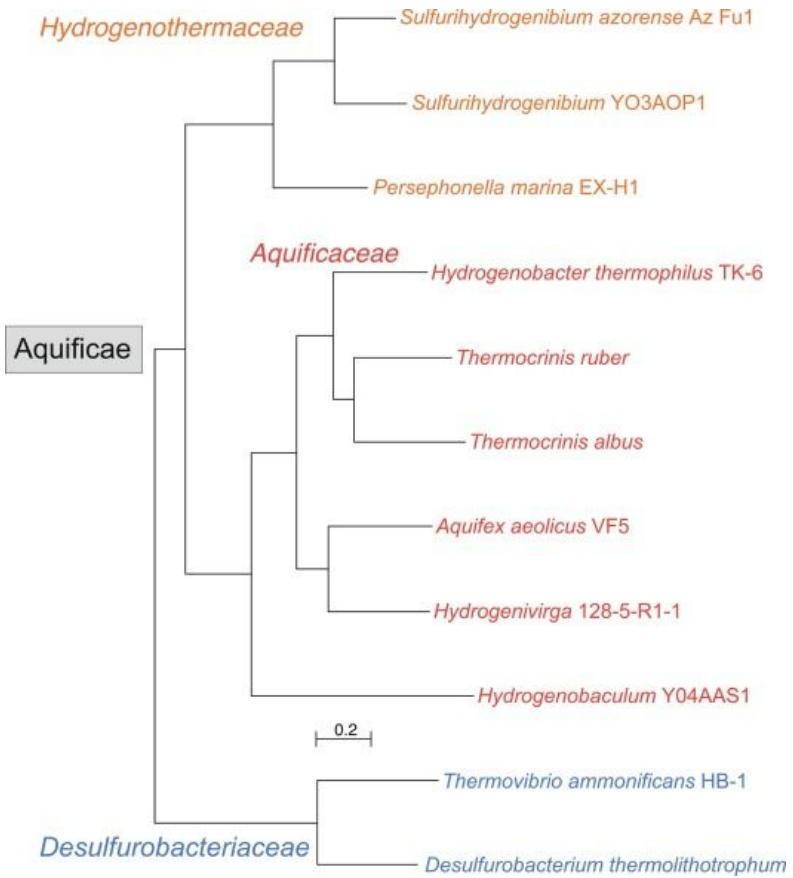
(a) **Major phyla of *Bacteria***

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Bacterial phyla: Aquificae

Aquificae are an early branching phylum of thermophilic and hyperthermophilic bacteria, mostly chemolithoautotrophic, found in both terrestrial and marine geothermal environments, such as deep-sea hydrothermal vents.

They generally use hydrogen as electron donor, fix CO_2 using the rTCA cycle and may use either nitrate, sulfur or low partial pressure of oxygen as electron acceptor.



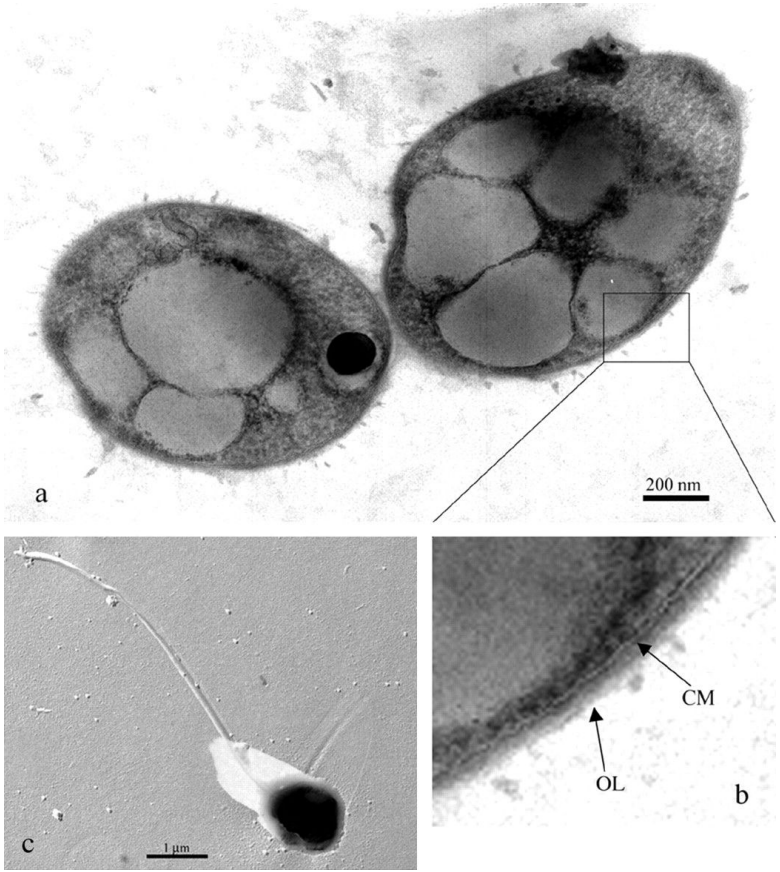
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The Aquificae *Themovibrio ammonifican* HB1

Thermovibrio ammonificans is a Gram-negative, strictly anaerobic bacterium belonging to the phylum Aquificae and the family Desulfurobacteriaceae.

It was **isolated from** the **walls of active**, high-temperature deep-sea hydrothermal vent chimneys (**black smokers**) on the East Pacific Rise at 9° 50′ N.

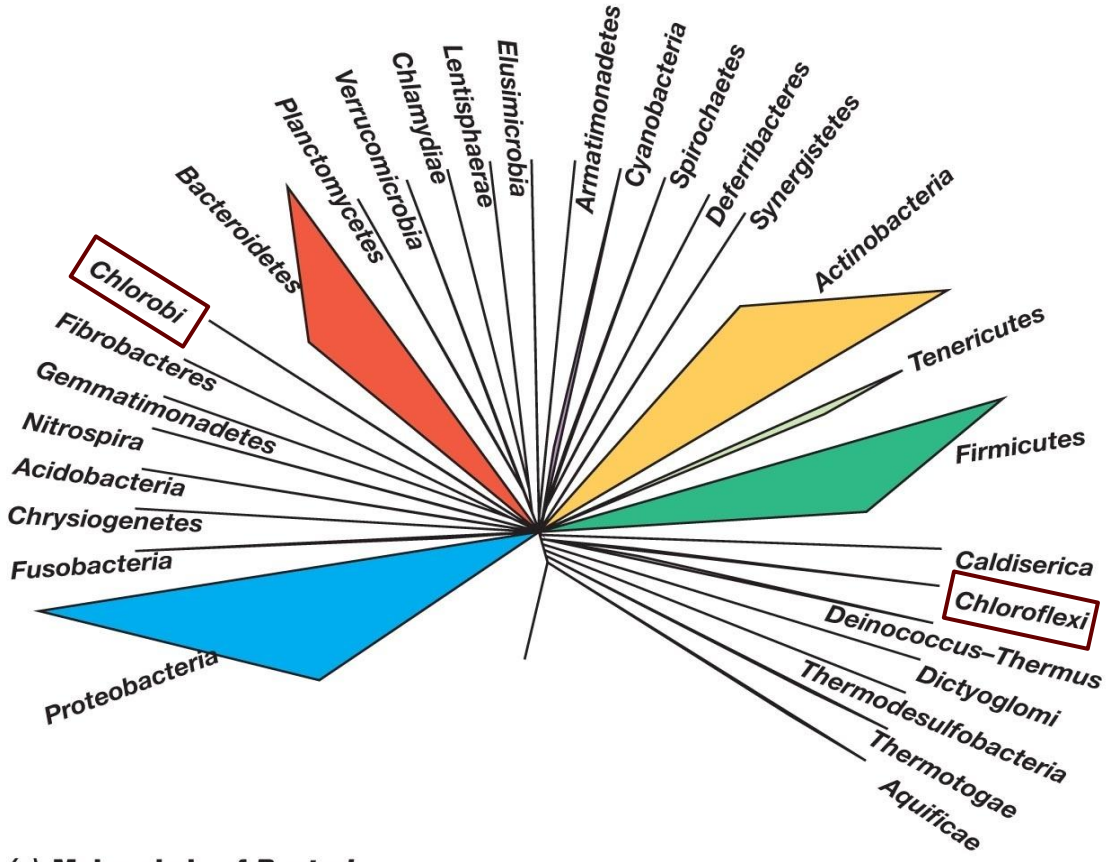
Growth occurred between 60 and 80°C (optimum at 75°C), under chemolithoautotrophic conditions in the presence of H_2 and CO_2, with nitrate or sulfur as the electron acceptor and with concomitant formation of ammonium or hydrogen sulfide, respectively.



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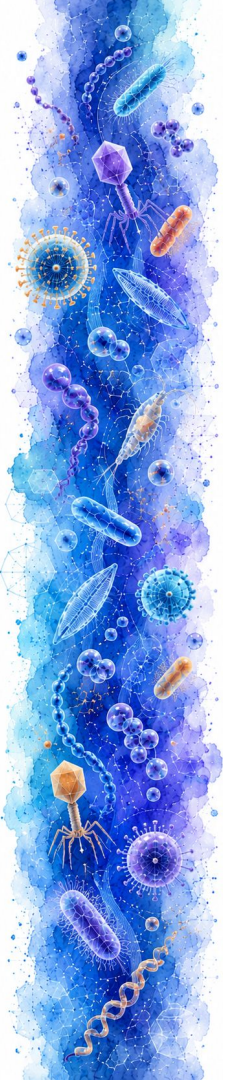
https://doi.org/10.1099/ijs.0.02781-0

The Tree Of Life: Bacteria



(a) **Major phyla of *Bacteria***

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Other Bacterial phyla

Chloroflexi are a phylum of bacteria containing isolates with a diversity of phenotypes including members that are aerobic thermophiles, anoxygenic photoheterotrophs (green non-sulfur bacteria), and anaerobic halorespirers, which uses halogenated organics (such as the toxic chlorinated ethenes and polychlorinated biphenyls) as energy sources. Although in the deep ocean they can comprise >40% of the total bacterial community, because of the absence of cultured marine representatives, their metabolic activity and role in marine processes are still unclear.

Chlorobi are mainly represented by the green sulfur bacteria, or Chlorobiales, a family of obligately anoxygenic photolithoautotrophic bacteria, found in shallow-water hydrothermal vents. They use light energy and reduced sulfur compounds as the electron source, but also hydrogen or ferrous iron can be used as electron donor. The major photosynthetic pigment in these bacteria is Bacteriochlorophylls c or d in green species and e in brown species, and is located in the chlorosomes and plasma membranes.

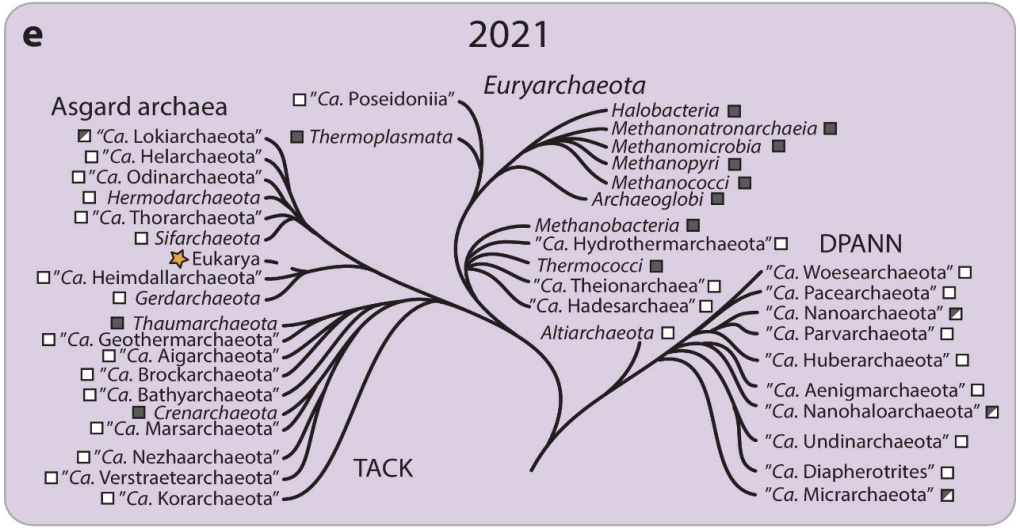
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Archaea

From 1990 through 2021, numerous archaeal phylum-level lineages and superphyla were discovered as well as re-positioned within the tree of life.

Crenarchaeota and **Euryarchaeota** are the first two archaeal phyla to be described. Then, the phylum **Nanoarchaeota** has been proposed in 2003 and the newly discovered **Thaumarchaeota** in 2008. Although the phylum **Korarchaeota** in 1996 by a team of scientists studying Archaea in hot springs located in Yellowstone National Park, it still lack cultivated representatives.

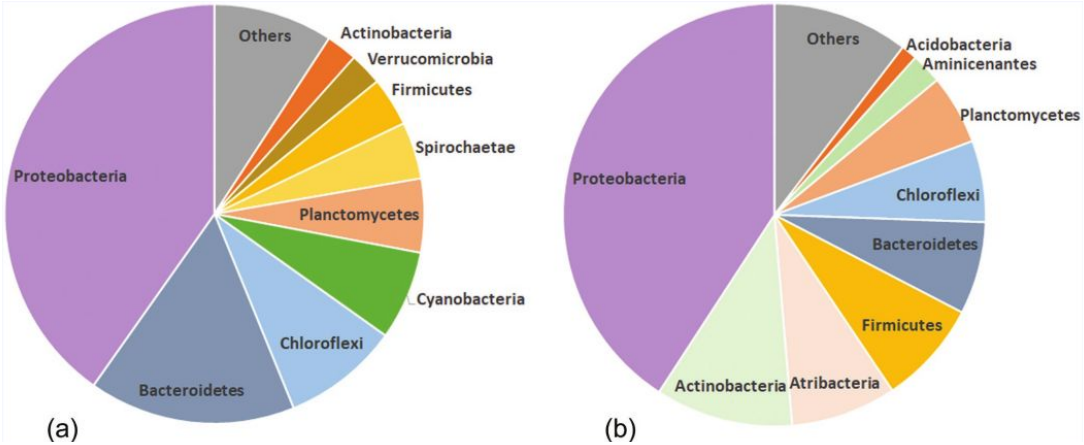


A **purple square** indicates that a **cultivated** representative of the phylum is deposited as a type strain to a public repository, a **half-purple square** indicates that a **(co)cultivated member** is described in the literature, and a **white square** indicates that **no cultivated** representative has been described.

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10.1146/annurev-micro-040921-050212

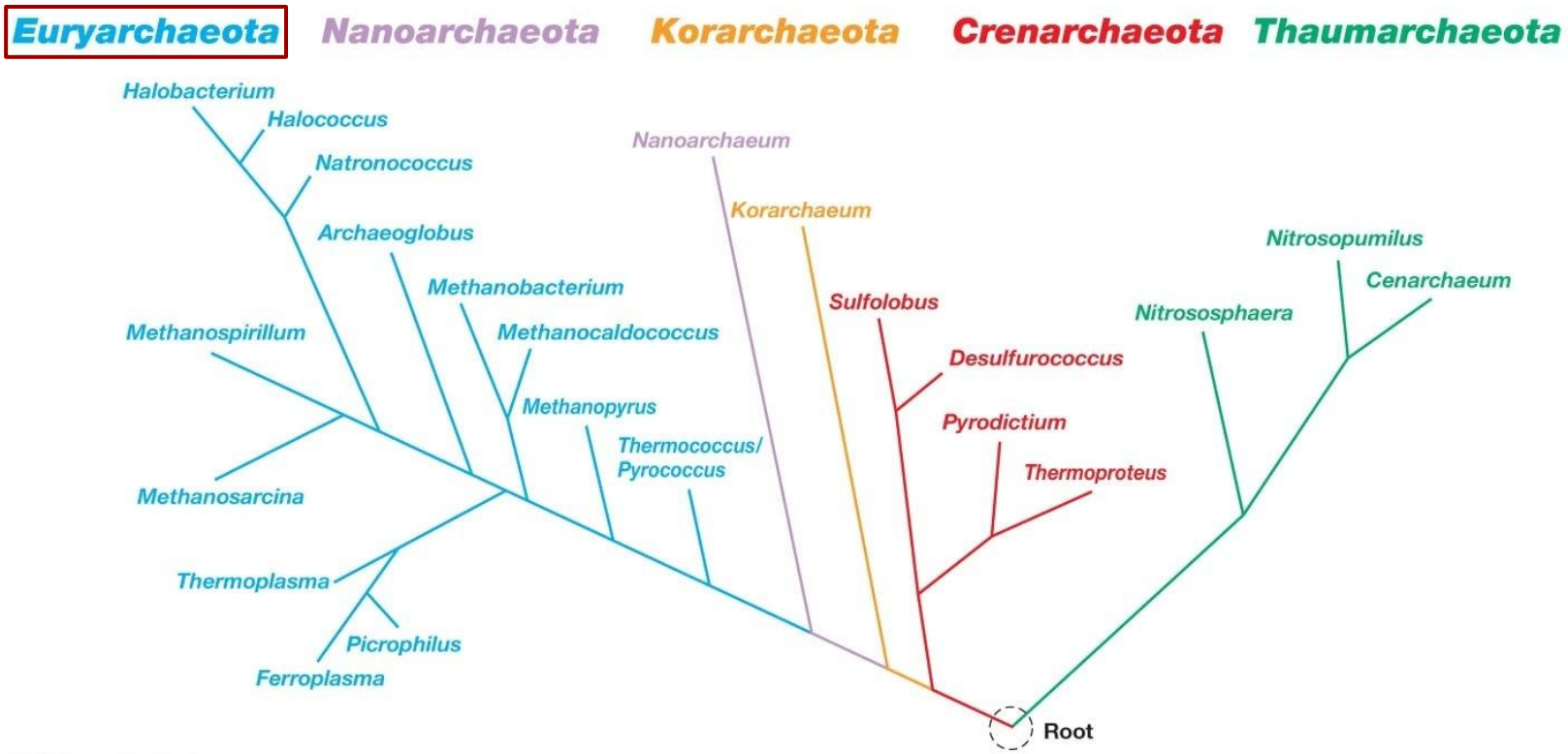
Bacteria and Archaea In Marine Environment



Frequency of sequences for bacterial phyla from **marine water samples (a)** and **marine sediments and hydrothermal vents (b)**, and for **archaeal phyla (c)**.

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The Tree Of Life: Archaea



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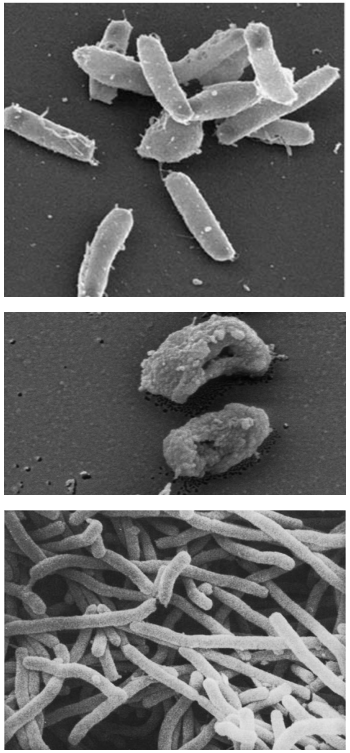
Archaeal phyla: Euryarchaeota

The Euryarchaeota (Greek for "broad old quality") are a phylum of the Archaea comprising a large variety of metabolic strategies. They include **methanogens**, **ANME**, **halophilic species** which survive extreme concentrations of salt, and some extremely thermophilic aerobes and anaerobes.

Hyperthermophiles are obligately anaerobic chemoorganotroph, using complex substrates such as proteins and carbohydrates, with sulfur as the electron acceptor, living in anoxic thermal waters at temperature ranging from 55 to 95 °C.

Methanogens are mesophilic or thermophilic obligate anaerobes and can be isolated using strict anaerobic techniques from a wide range of habitats including anoxic sediments, decomposing material, and the gut of animals.

Extreme halophiles use amino acids or organic acids as electron donors aerobically and require a number of growth factors such as vitamins for optimal growth, living at salinities ranging from 2 to 5.5 M NaCl (32%, the limit of saturation for NaCl).

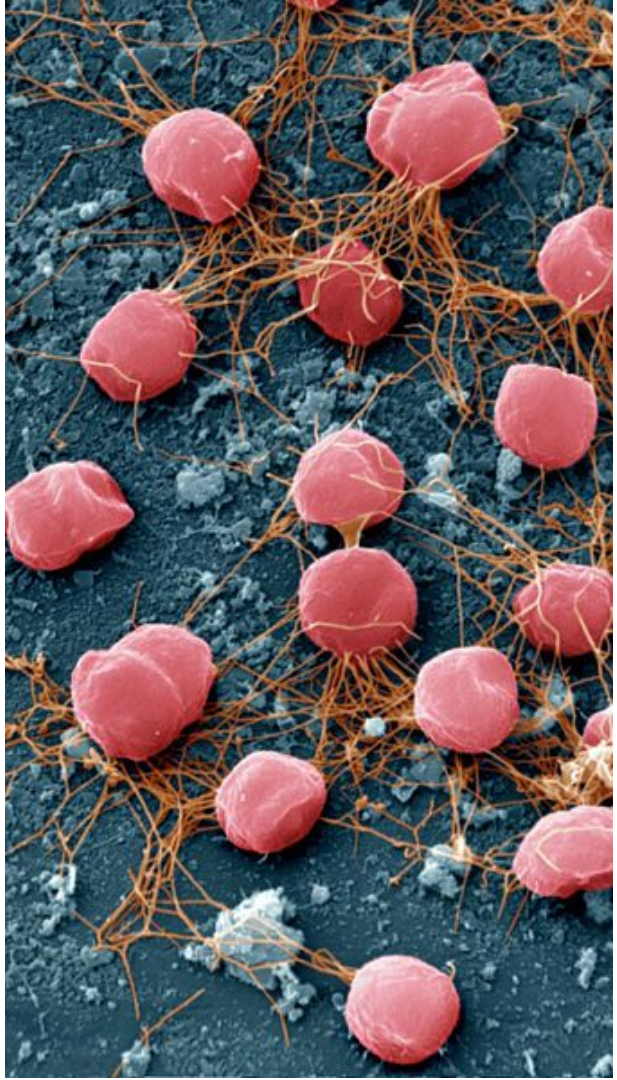


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Archaeal phyla: Euryarchaeota

Pyrococcus furiosus is a strictly anaerobic and heterotrophic archaeon, first isolated from a hydrothermal vent near Vulcano Island, Italy.
It grows well on complex organic mixtures (yeast extract, maltose, starch) and protein sources, and uses S^0 as terminal electron acceptor, generating H_2S when S^0 is present, or and only H_2 in non sulfur environments.

P. furiosus cells appears as mostly regular cocci of 0.8 µm to 1.5 µm diameter with monopolar polytrichous flagellation. It is hyperthermophilic, usually growing at temperatures ranging from 70 to 106 °C, with an optimum at 100 °C, and between 5-9 pH, with an optimum at 7.



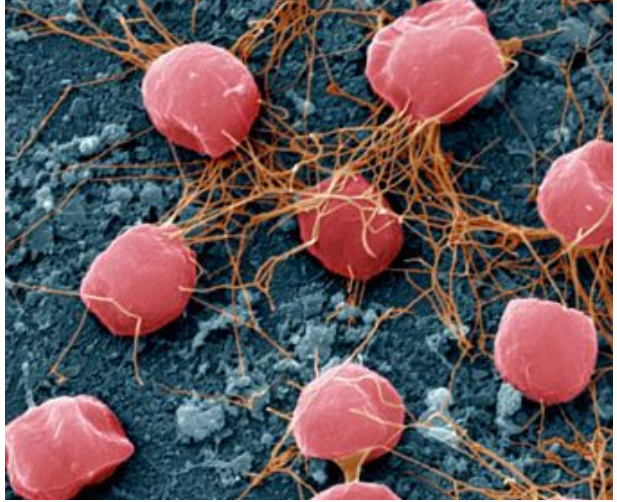
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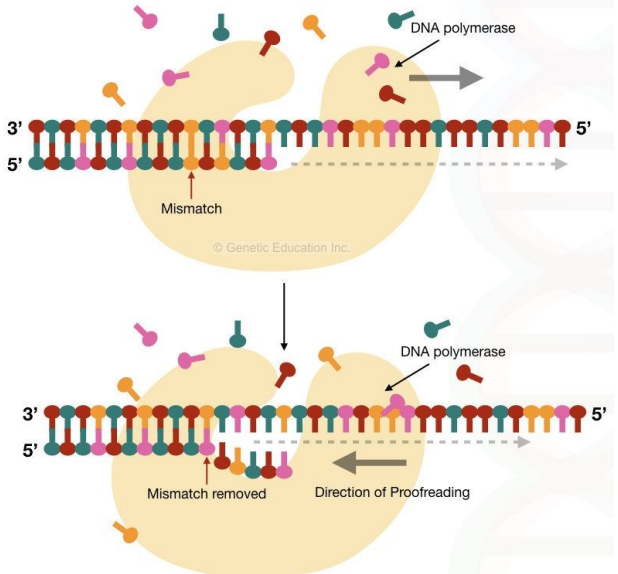
Archaeal phyla: Euryarchaeota

Pyrococcus furiosus is a strictly anaerobic and heterotrophic archaeon, first isolated from a hydrothermal vent near Vulcano Island, Italy.
It grows well on complex organic mixtures (yeast extract, maltose, starch) and protein sources, and uses S^0 as terminal electron acceptor, generating H_2S when S^0 is present, or and only H_2 in non sulfur environments.

P. furiosus is one of the few organisms identified as possessing aldehyde ferredoxin oxidoreductase, enzyme involved in the reduction of organic acids to their corresponding alcohol, containing tungsten, an element rarely found in biological molecules.

Pfu DNA polymerase isolated from *P. furiosus*, is now widely used due to its superior thermostability and proofreading properties compared with Taq DNA polymerase.



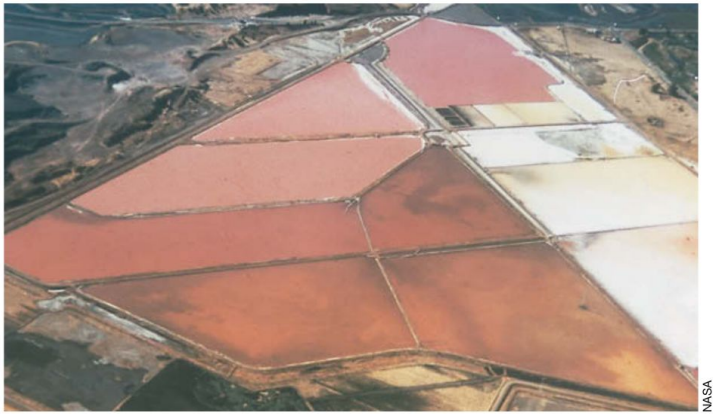


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Archaeal phyla: Euryarchaeota - Halophilic

Table 16.1 Some genera of extremely halophilic *Archaea*

Genus	*Morphology*	*Habitat*
Extreme halophiles		
Halobacterium	Rods	Salted fish; hides; hypersaline lakes; salterns
Halorubrum	Rods	Dead Sea; salterns
Halobaculum	Rods	Dead Sea
Haloferax	Flattened discs	Dead Sea; salterns
Haloarcula	Irregular discs	Salt pools, Death Valley, CA; marine salterns
Halococcus	Cocci	Salted fish; salterns
Halogeometricum	Pleomorphic flat cells	Solar salterns
Haloterrigena	Rods, ovals	Saline soil
Haloquadratum	Flat squares	Salterns
Haloalkaliphiles		
Natronobacterium	Rods	Highly saline soda lakes
Natrinema	Rods	Salted fish; hides
Natrialba	Rods	Soda lakes; beach sand
Natronomonas	Rods	Soda lakes
Natronococcus	Cocci	Soda lakes
Natronorubrum	Flattened cells	Soda lakes

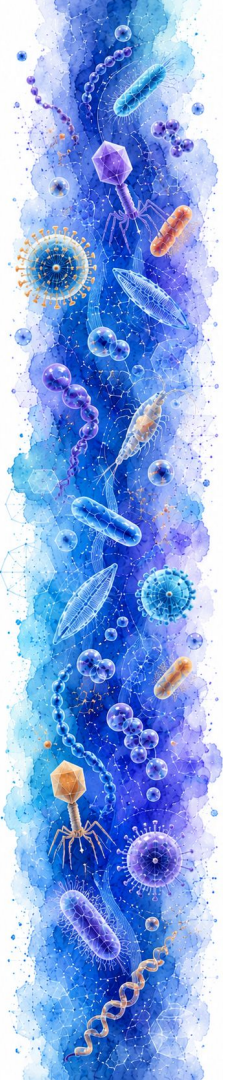


NASA



Michael T. Madigan

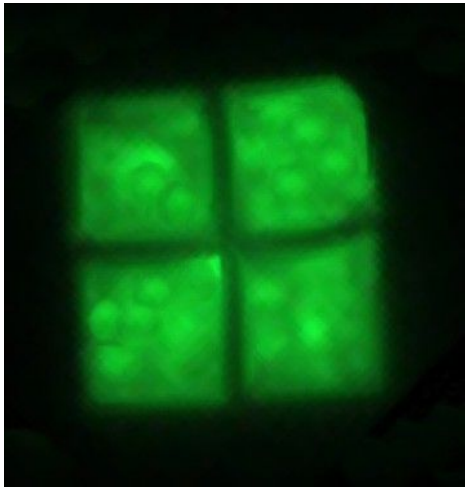
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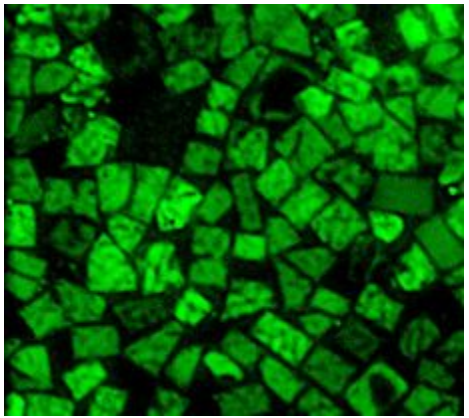


Archaeal phyla: Euryarchaeota

Haloquadratum ("salt square") is a genus of the family Halobacteriaceae containing a single genus and a single species, *Haloquadratum walsbyi*, a phototrophic halophilic archaeon. Discovered in 1980 in the Gavish Sabkha, a coastal brine pool on the Sinai Peninsula in Egypt, the archaeon was not cultured until 2004.

H. walsbyi can be found anywhere in hypersaline waters. When sea water evaporates, calcium carbonate and calcium sulfate precipitate, leading to a hypersaline sodium chloride-rich brine. Further evaporation results in the precipitation of sodium chloride or halite, and then to a concentrated magnesium chloride brine termed **bittern**. During the final stage of halite formation, before magnesium chloride concentration causes the brine to become sterile, *H. walsbyi* flourishes and can make up 80% of the brine's biomass. The cells typically contain PHA granules (polyhydroxyalkanoates), and a large numbers of gas-filled vacuoles which provide buoyancy and may help to position the cells to maximise light-harvesting





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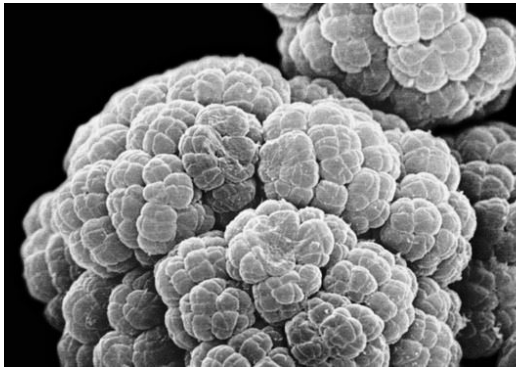
Archaeal phyla: Euryarchaeota - Methanogens

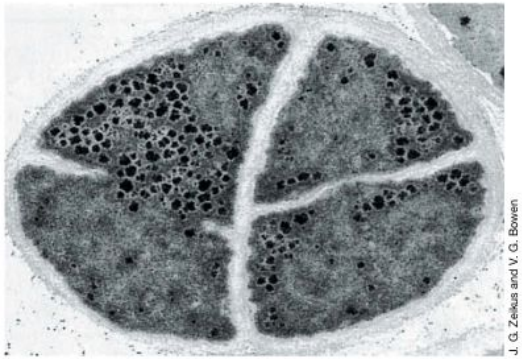
Methanogens are coccoid or bacilli shaped archaea belonging to the Euryarchaeota phylum. There are over 50 described species of methanogens, which do not form a monophyletic group, although all methanogens belong to Archaea

They are strict anaerobic organisms, typically living in environments in which all electron acceptors other than CO_2 have been depleted. They have been found in several extreme environments on Earth, mostly abundant in deep subterranean habitats.

They play a key ecological role in anaerobic environments **removing** excess hydrogen, used as electron donor, and fermentation products and contributing to organic matter decay.

Most methanogens are autotrophic producers, but **those that oxidize CH_3COO^- are classified as chemotroph instead**. Closely related to the methanogens are the anaerobic methane oxidizers (ANME), often inhabiting the same niches.





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Archaeal phyla: Euryarchaeota - Methanogens

Table 16.3 Habitats of methanogens

I. Anoxic sediments: marsh, swamp, and lake sediments, paddy fields, moist landfills
II. Animal digestive tracts:[a]
  A. Rumen of ruminant animals such as cattle, sheep, elk, deer, and camels
  B. Cecum of cecal animals such as horses and rabbits
  C. Large intestine of monogastric animals such as humans, swine, and dogs
  D. Hindgut of cellulolytic insects (for example, termites)
III. Geothermal sources of H_2 + CO_2: hydrothermal vents
IV. Artificial biodegradation facilities: sewage sludge digesters
V. Endosymbionts of various anaerobic protozoa

Table 16.5 Substrates converted to methane by various methanogenic *Archaea*

I. CO_2-type substrates
Carbon dioxide, CO_2 (with electrons derived from H_2, certain alcohols, or pyruvate)
Formate, $HCOO^-$
Carbon monoxide, CO

II. Methylated substrates
Methanol, CH_3OH
Methylamine, $CH_3NH_3^+$
Dimethylamine, $(CH_3)_2NH_2^+$
Trimethylamine, $(CH_3)_3NH^+$
Methylmercaptan, CH_3SH
Dimethylsulfide, $(CH_3)_2S$

III. Acetotrophic substrates
Acetate, CH_3COO^-
Pyruvate, CH_3COCOO^-

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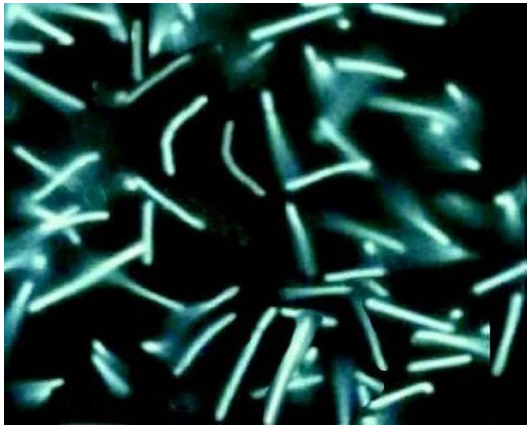
Archaeal phyla: Euryarchaeota - Methanogens

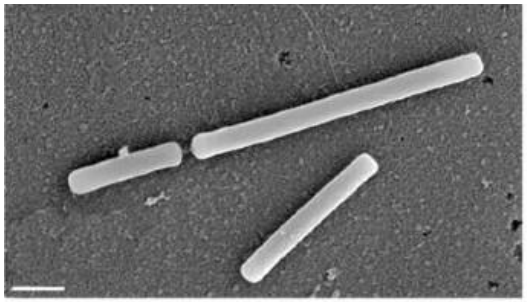
Methanopyrus kandleri is rod-shaped methanogen euryarchaeota, obligate anaerobic chemolithoautotroph. Cells have approximate length of 2-14 μm and diameter of 0.5 μm

It utilizes hydrogen as an electron source and reduces carbon dioxide from the environment into methane.
It is a hyperthermophilic archaeon, usually growing at temperatures ranging from 84 to 110°C, with an optimum at 98°C, and between 5.5-7 pH, with an optimum at 6.5.

M. kandleri was first isolated from the wall of "black smoker" hydrothermal vent chimneys from the Gulf of California at a depth of 2000 m.

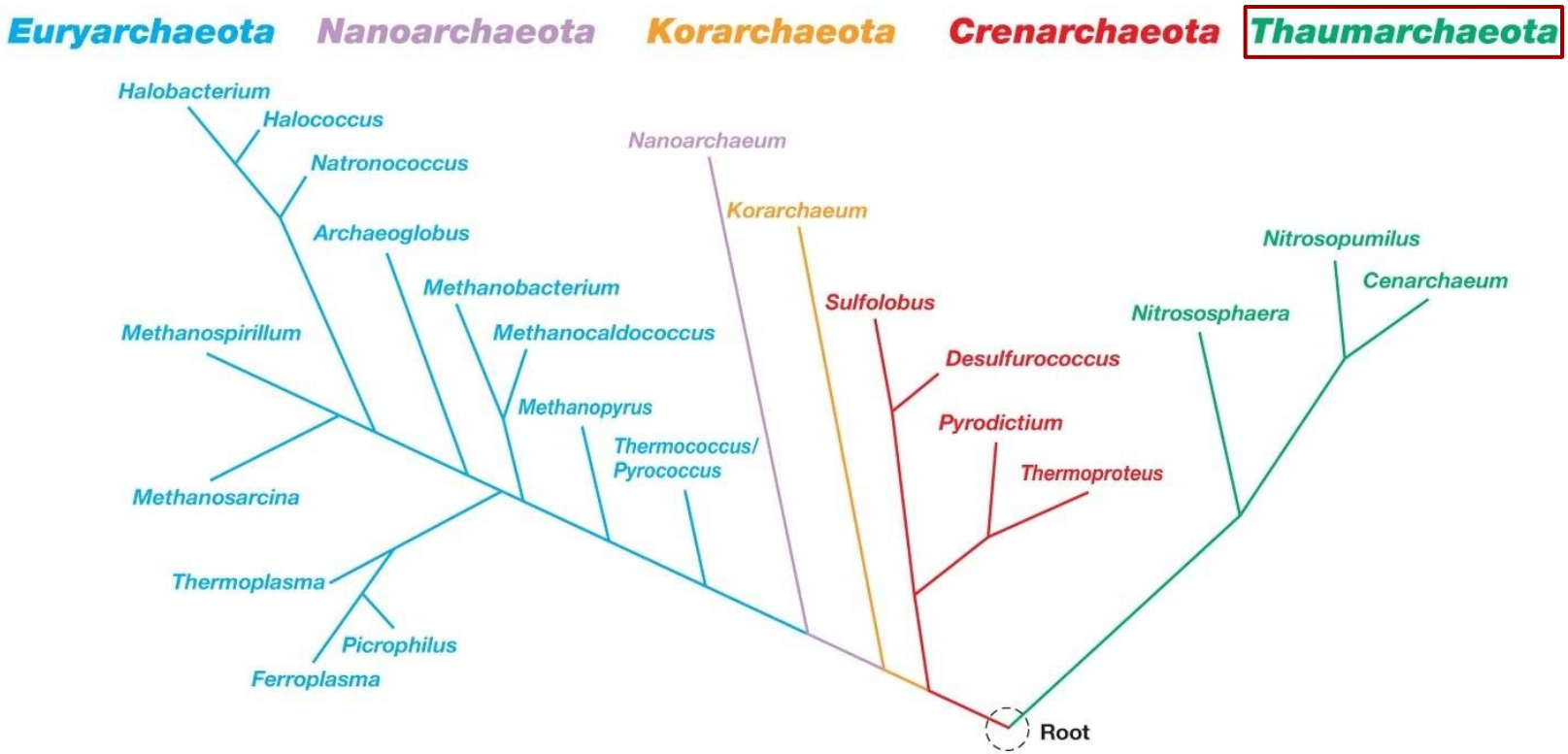
Strain 116 was reported to grow at 122 °C, the highest temperature yet shown to support microbial growth, in particular pressure conditions.





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The Tree Of Life: Archaea



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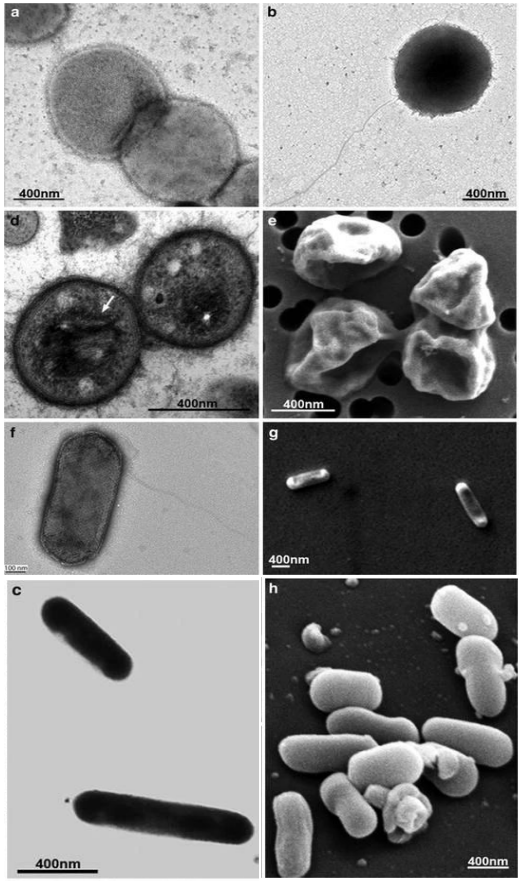
Archaeal phyla: Thaumarchaeota

The Thaumarchaeota (from the Greek 'thaumas', meaning wonder) are a phylum of the Archaea proposed in 2008 after the genome of Cenarchaeum symbiosum (the first isolate of this group) was sequenced and found to differ significantly from other members of the hyperthermophilic phylum Crenarchaeota.

Sequences belonging to this phylum have been identified in numerous marine and soil environments and are now believed to be **ubiquitous**.

All organisms of this lineage so far identified are chemolithoautotrophic ammonia-oxidizers and may play important roles in biogeochemical cycles, such as the nitrogen cycle and the carbon cycle.

Only few species have been isolated in addition to *C. symbosium*: *Nitrosopumilus maritimus*, *Nitrososphaera viennensis*, and *Nitrososphaera gargensis*



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10.1007/978-3-642-38954-2_338

Archaeal phyla: Thaumarchaeota

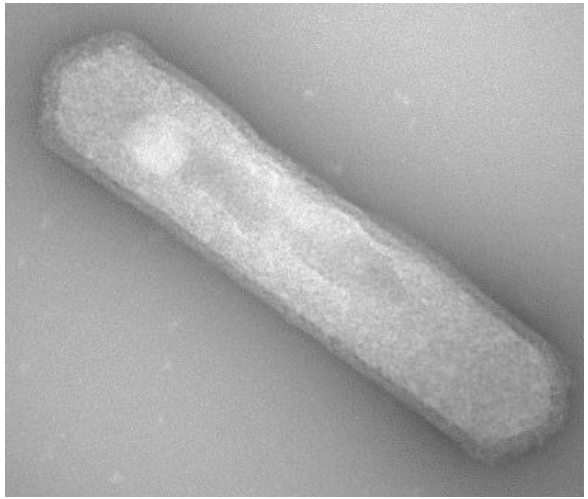
Nitrosopumilus maritimus is a member of the newly proposed archaea phylum Thaumarchaeota.

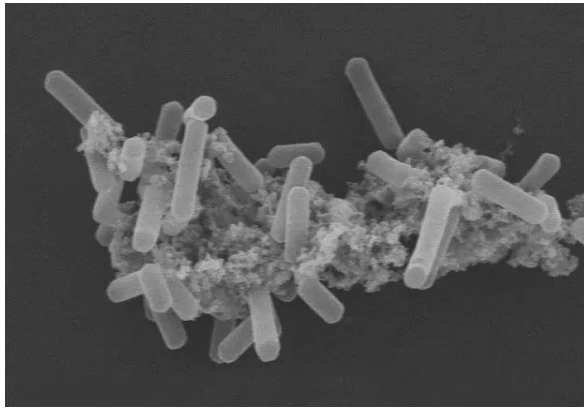
N. maritimus was the first **non-thermophilic** archaeon to be maintained in axenic culture. The original isolate was obtained from a tropical salt-water aquarium.

Archaeal cells (mostly members of the phylum now known as Thaumarchaeota) comprise a large fraction of the picoplankton, especially in deeper waters.

It was the first reported chemolithoautotroph nitrifier in the domain archaea.

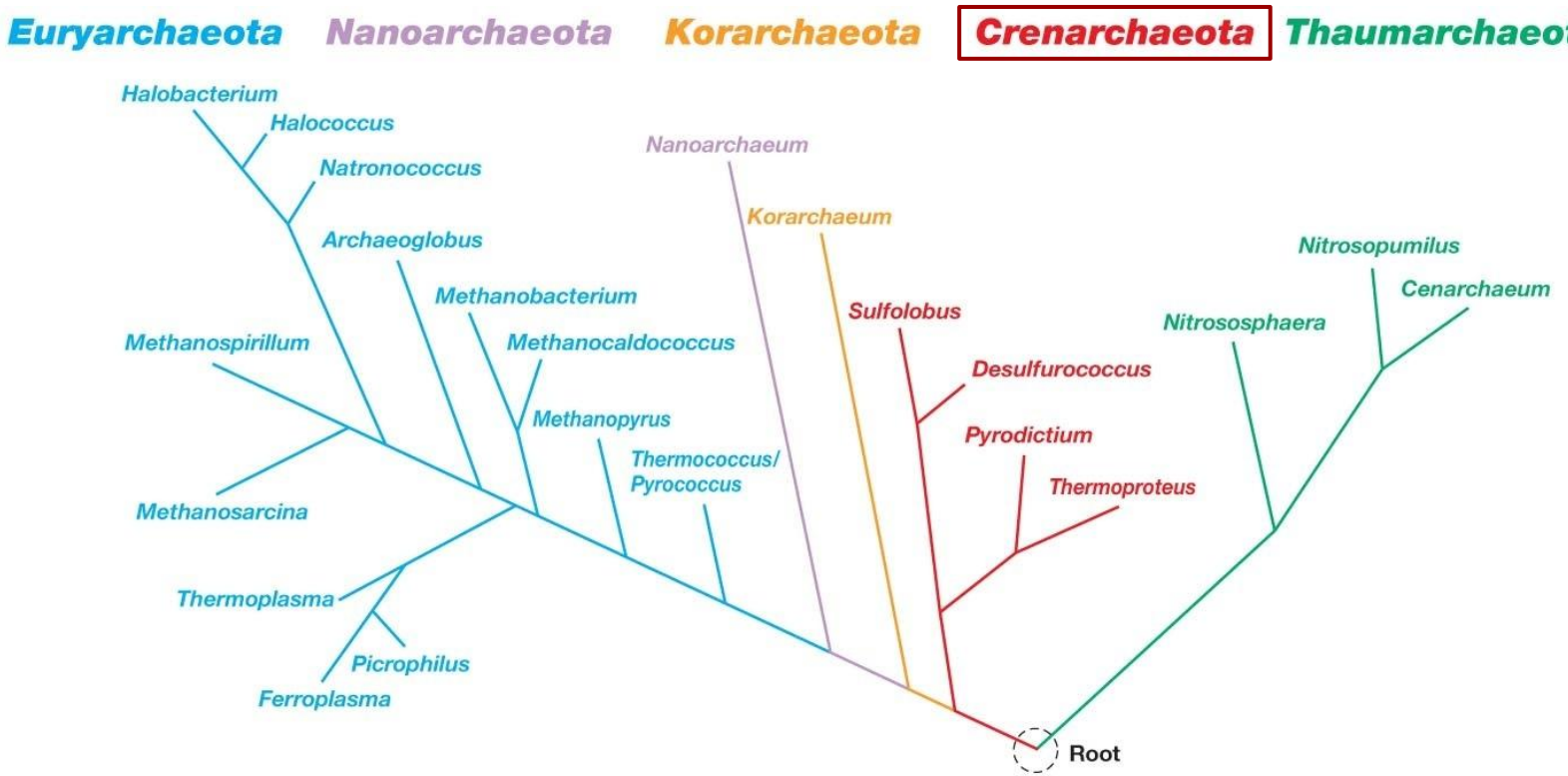
It is believed to be a key player in marine nitrification. This hypothesis is supported by the parallels between the depth of the marine nitrite maximum and the vertical distribution of Thaumarchaeota in the ocean.



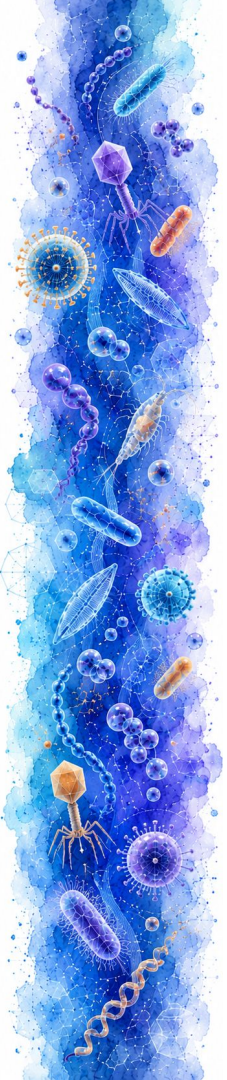


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The Tree Of Life: Archaea



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Archaeal phyla: Crenarchaeota

The Crenarchaeota, initially thought to be composed only by acidophilic sulfur-dependent extremophiles, account approximately 15 known genera. Most of the hyperthermophilic species have been isolated from marine or terrestrial volcanic environments, such as sulfur-rich environments, called solfataras, and shallow or deep-sea hydrothermal vents.

Recent analyses of genetic sequences obtained directly from environmental samples, however, indicate the existence of low temperature Crenarchaeota, which have not yet been cultivated.

Cells range in size from cocci <1 μm in diameter to filaments over 100 μm in length. Species display a wide range of cell shapes, including regular cocci clustered in grape-like aggregates. (Staphylothermus), irregular, lobed cells (Sulfolobus), discs (Thermodiscus), very thin filaments (<0.5 μm diameter; Thermofilum), and almost rectangular rods (Thermoproteus, Pyrobaculum). Most species possess flagella and are motile.

Metabolically, Crenarchaeota are quite diverse, ranging from chemoorganotrophs to chemolithoautotrophs. They are anaerobes, facultative anaerobes or aerobes, and many utilize sulfur for energy metabolism

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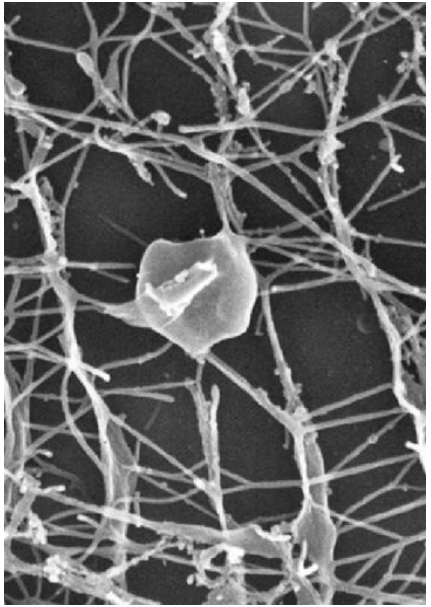
Crenarchaeota: *Pyrodictium abyssi*

Pyrodictium abyssi is a species of submarine strict anaerobe hyperthermophilic archaea, that grows chemolithotrophically on H_2 as an electron donor and S^0 as an electron acceptor, but can also grow chemoorganotrophically

Pyrodictium are generally found in the porous walls of deep-sea vents where the inside temperatures get as high as 400 °C, but usually growing at temperatures ranging from 80 to 110 °C, with an optimum at 105 °C.

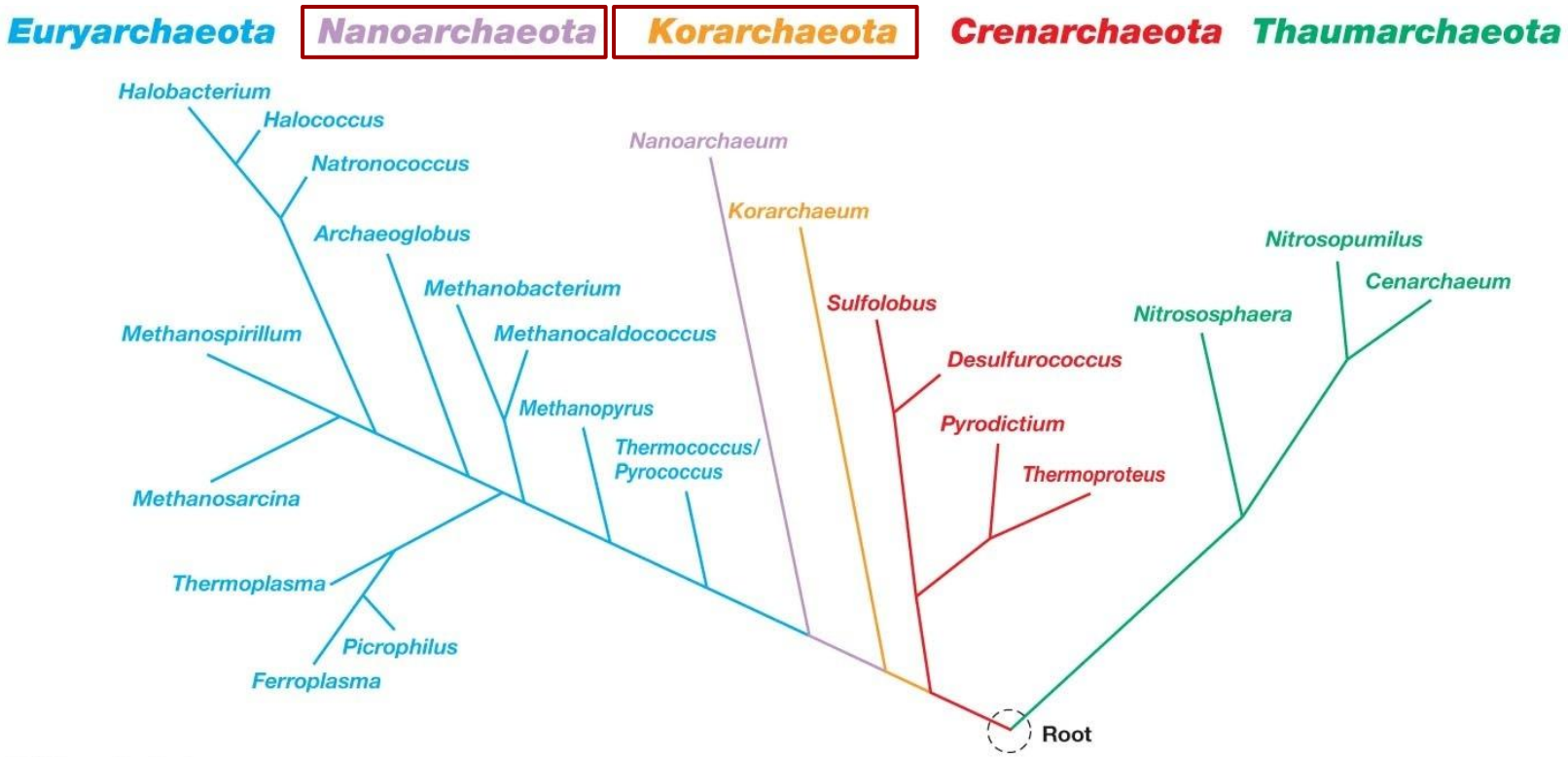
Pyrodictium cells appear as flat disks, with a large size range (300 to 2500 nanometres in diameter) which may allow Pyrodictium to inhabit a variety of pores

The cells grow in unique flake-like shapes held together by a network of hollow cannulae (tubules). The cannulae branch out and connect with other cells. It is likely that the range of motion provided by the cannulae allow the cells to move freely thanks to thermal energy from the extreme heat of the deep-sea vents.



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The Tree Of Life: Archaea

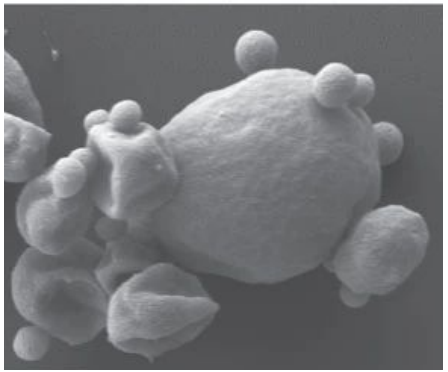


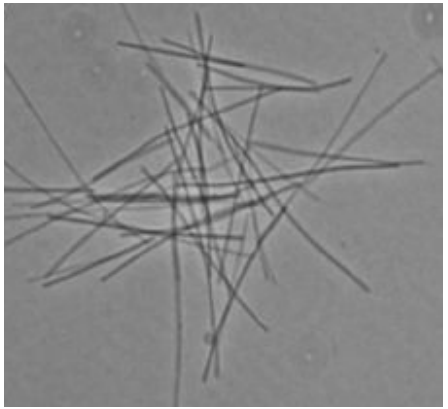
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Archaeal phyla: Nanoarchaeota and Korarchaeota

Nanoarchaeota (Greek, "dwarf or tiny ancient one") are a phylum of the small Archaea. This phylum currently has only one representative, Nanoarchaeum equitans. They are found living attached to other archaea cells and are believed to be parasites.

Korarchaeota (Greek, "young ancient man") are a newly proposed phylum of the Archaea. Analysis of their 16S rRNA gene sequences suggests that they are a deeply branching lineage. They have only been found in hydrothermal environments.





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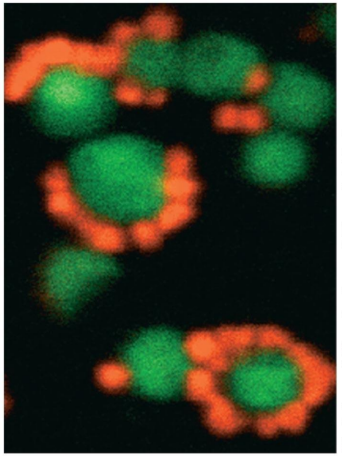
Archaeal phyla: Nanoarchaeota and Korarchaeota

Nanoarchaeum equitans is a species of marine Archaea "parasite" that was discovered in 2002 in a hydrothermal vent off the coast of Iceland

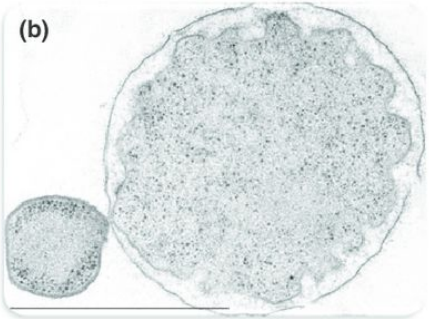
Nanoarchaeum appears to be an obligate symbiont on the archaeon Ignicoccus; it must be in contact with the host organism to survive. It cannot synthesize lipids but obtains them from its host. Whether it obtains energy from biological molecules imported from Ignicoccus, or whether it receives ATP directly is currently unknown.

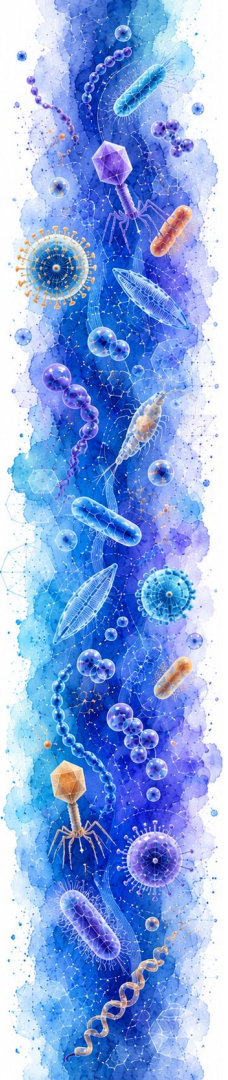
Its cells are only 400 nm in diameter, making it one of the smallest known cellular organisms, and the smallest known archaeon.

N. equitans genome consists of a single circular chromosome of 0.49 Mbp, coding for 540 genes.



Reinhard Rachel and Harald Huber





New Taxonomy Nomenclature

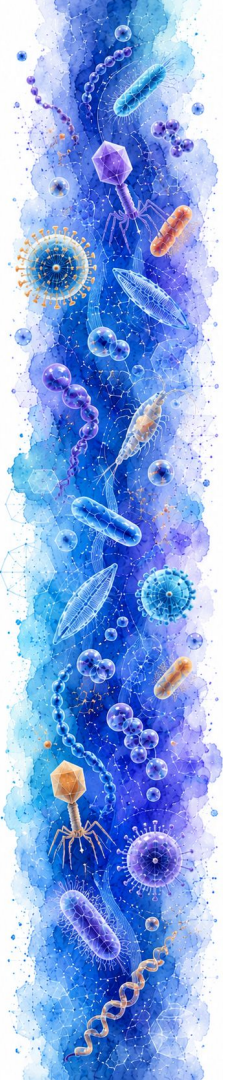
Since January 2023, NCBI Taxonomy prokaryote phylum names are changed in accordance with the recent inclusion of rank 'phylum' in the International Code of Nomenclature for Prokaryotes (ICNP).

The full list of changes:

Taxid	Old Name	New Name	Note
976	**Bacteroidetes**	**Bacteroidota**	
203682	**Planctomycetes**	**Planctomycetota**	
57723	Acidobacteria	Acidobacteriota	
201174	**Actinobacteria**	**Actinomycetota**	
200783	**Aquificae**	**Aquificota**	
67819	Armatimonadetes	Armatimonadota	
1936987	Balneolaeota	Balneolota	
67814	Caldiseric	Caldisericota	
1930617	Calditrichaeota	Calditrichota	
204428	Chlamydiae	Chlamydiota	
1090	Chlorobi	Chlorobiota	
200795	Chloroflexi	Chloroflexota	
200938	Chrysiogenetes	Chrysiogenota	
1117	**Cyanobacteria**	**Cyanobacteriota**	
28889	**Crenarchaeota**	**Thermoproteota**	
200930	Deferribacteres	Deferribacterota	
1297	Deinococcus-Thermus	Deinococcota	
68297	Dictyoglomi	Dictyoglomota	
74152	Elusimicrobia	Elusimicrobiota	
65842	Fibrobacteres	Fibrobacterota	
1239	**Firmicutes**	**Bacillota**	
32066	Fusobacteria	Fusobacteriota	
142182	Gemmatimonadetes	Gemmatimonadota	
1134404	Ignavibacteriae	Ignavibacteriota	
134625	Kiritimatiellaeota	Kiritimatiellota	
256845	Lentisphaerae	Lentisphaerota	
1293497	Nitrospinae	Nitrospinota	
40117	**Nitrospirae**	**Nitrospirota**	
1224	**Proteobacteria**	**Pseudomonadota**	
1853220	Rhodothermaeota	Rhodothermota	
203691	Spirochaetes	Spirochaetota	
508458	Synergistetes	Synergistota	
544448	Tenericutes	Mycoplasmatota	
651137	**Thaumarchaeota**	**Nitrososphaerota**	
67799	Thermodesulfobacteria	Thermodesulfobacteriota	
200918	**Thermotogae**	**Thermotogota**	
74201	Verrucomicrobia	Verrucomicrobiota	
2818505	Myxococcota		(New Phylum)
29547	**Epsilonproteobacteria**	**Campylobacterota**	**[Class → Phylum]**
189775	Thermomicrobia	Thermomicrobiota	[Class → Phylum]
1553900	Oligoflexia	Bdellovibrionota	[Class → Phylum]
28221	**Deltaproteobacteria**	**Desulfobacterota**	**[Class → Phylum]**

https://ncbiinsights.ncbi.nlm.nih.gov/2022/11/14/prokaryotic-phylum-name-changes/

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This Week Read

Baker, Brett J., Valerie De Anda, Kiley W. Seitz, Nina Dombrowski, Alyson E. Santoro, e Karen G. Lloyd. 2020. «Diversity, Ecology and Evolution of Archaea». Nature Microbiology 5 (7): 887–900. https://doi.org/10.1038/s41564-020-0715-z